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Evaluation of stochastic trajectory-based epidemic models using the energy score --Manuscript Draft--

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Epidemics
Editorial Board

Dear Editors,

We are pleased to submit our manuscript titled “Evaluation of stochastic trajectory-based epidemic models using the energy score” by Bay et al. for consideration in *Epidemics*.

The importance of rigorous evaluation frameworks for epidemic forecasts and scenario projections has grown significantly as multi-team collaborative efforts, such as the Scenario Modeling Hub and the CDC’s FluSight Challenge, have become essential for informing public health decisions. This work introduces the energy score as a well-suited performance metric for evaluating stochastic trajectory-based epidemic models, addressing limitations in current scoring methodologies for multivariate time-series epidemic predictions.

In this paper, we present the energy score, a multivariate extension of the continuous ranked probability score (CRPS), for evaluating epidemic projections in stochastic trajectory format. More specifically we provide the following contributions:

- We compare the energy score with WIS, demonstrating its ability to distinguish between forecasts that have the same marginal distribution but differ in trajectory structure.
- We illustrate how the energy score improves evaluation of complex, multi-peak epidemic models by overcoming the limitations of model communication using fixed-time descriptive statistics (e.g. quantiles) in capturing the full variability of model trajectories.
- We apply the energy score to the 2023–2024 Flu Scenario Modeling Hub projections, illustrating how this metric performs across multiple scenarios and models submitted by various teams.
- We introduce a normalized energy score to assess multi-target outcomes (e.g., cases, hospitalizations, deaths, geographical locations) in a unified metric, improving interpretability for public health decision-making.

We believe these contributions are timely, given the increasing adoption of ensemble epidemic forecasts and the need for evaluation frameworks that accommodate trajectory-based epidemic models. Our study provides practical insights for improving predictive analytics in public health, with implications for epidemic forecasting and scenario modeling hubs. For this reason, we view *Epidemics* as the ideal venue to share these findings and engage the scientific community most invested in methodological innovation for epidemic modeling and evaluation.

Thank you very much for your consideration.

Sincerely, on behalf of all the authors,

Clara Bay

Evaluation of stochastic trajectory-based epidemic models using the energy score

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Abstract

Scoring rules are critical for evaluating the predictive performance of epidemic models by quantifying how well their projections and forecasts align with observed data. In this study, we introduce the energy score as a performance metric for stochastic trajectory-based epidemic models. As a multivariate extension of the continuous ranked probability score (CRPS), the energy score provides a single, unified measure for time-series predictions. It evaluates both calibration and sharpness by considering the distances between individual trajectories and observed data, as well as the inter-trajectory variability. We provide an overview of how the energy score can be applied to assess both scenario projections and forecasts in this format, with a particular focus on a detailed analysis of the Scenario Modeling Hub results for the 2023-2024 influenza season. By comparing the energy score to the widely used weighted interval score (WIS), we demonstrate its utility as a tool for evaluating epidemic models, especially in scenarios requiring integration of predictions across multiple target outcomes into a single, interpretable metric.

Introduction

Epidemic models often generate probabilistic predictions, representing a range of plausible future trajectories rather than a single outcome. In contrast to deterministic forecasts based on fixed initial conditions, these probabilistic outputs incorporate uncertainty arising from random processes and variable parameter values (Sherratt et al., 2024). Clear communication of such uncertainty is essential for informing decision-makers about the spectrum of possible epidemic scenarios (McCabe et al., 2021).

Epidemic forecasting and scenario modeling initiatives, such as the CDC FluSight Forecasting Challenge, the COVID-19 Forecast Hub, and the Scenario Modeling Hub (SMH), require participating teams to submit probabilistic predictions in a quantile format. (Mathis et al., 2024; Cramer et al., 2022a; Howerton et al., 2023). Scoring rules, such as the weighted interval score (WIS), can then be applied to quantile format outputs to analyze the performance of projections with respect to observed surveillance data. Epidemic predictions, however, are often produced as collections of stochastic trajectories, each representing one possible

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realization of how an outbreak could evolve(Sherratt et al., 2024). Increasingly, collaborative forecasting hubs have requested these individual trajectories—rather than only aggregated quantile summaries—to provide greater transparency and a clearer view of the predicted epidemic dynamics. This shift, exemplified in recent rounds of the Scenario Modeling Hub, helps avoid the opacity of prediction intervals, which can mask important temporal features of the modeled outcomes(Loo et al., 2024).

The information contained within epidemic projections in a quantile format versus trajectory format has both advantages and disadvantages. Trajectories allow for greater flexibility in weighting and analysis of time-series features, where the quantile format does not retain information on variability within potential epidemic outcomes (Sherratt et al., 2024). Trajectories acknowledge the individual behavior of multiple potential realizations of an outbreak, where the quantiles evaluate the descriptive statistics at each time point. Fixed-time descriptive statistics can suppress important epidemiological information and obscure the true uncertainty of different epidemic outcomes (Juul et al., 2021; McCabe et al., 2021; Sherratt et al., 2024). Typically, predictions in a quantile format highlight the most likely outcomes, rather than the worst-case scenario, which is critically important to public health decision-makers (Wilke and Bergstrom, 2020). While this has implications for the communication of epidemic projections, it can also impact what we can learn from model evaluation.

Although the value of trajectory-based outputs is increasingly recognized, most evaluation methods for epidemic projections are still designed for quantile-based formats and have not been widely adapted to leverage the richer information trajectories provide. Here, we assess the utility of the energy score as a performance metric specifically tailored to evaluate epidemic model projections reported in stochastic trajectory format. The energy score has been applied across various fields, including weather forecasting (Gneiting et al., 2008; Kapoor et al., 2023), electricity market pricing (Cramer et al., 2023; Grothe et al., 2023), and wind/solar power generation (Pinson and Girard, 2012; Staid et al., 2017; Zhang et al., 2014; van der Meer, 2021). In epidemic modeling, the energy score has been applied in a limited number of cases to analyze multivariate time-series models (Held et al., 2017; Engebretsen et al., 2023; Bonacina et al., 2024). Moreover, computational packages used to assess probabilistic forecasts with proper scores have implemented the energy score (Jordan et al., 2019; Allen, 2024; Zanetta and Allen, 2024). However, it has yet to gain widespread adoption as a standard metric for evaluating epidemic forecasts and projections reported in a trajectory format.

In this study, we define the energy score, outline methods to adapt it for specific applications, and

perform synthetic experiments to explore its properties. We then apply the energy score to evaluate the performance of models contributing to the 2023–24 Flu Scenario Modeling Hub (SMH). In the Flu SMH, modeling teams provide predictions about future influenza trajectories under certain assumptions about future human behavior, environmental factors, and circulating strains. The SMH has performed 5 cycles of influenza scenario projections; 3 during the 2022-23 influenza season and 1 for each of the 2023-24 and 2024-25 seasons in addition to 19 scenario projections of COVID-19 and 2 of RSV (Scenario Modeling Hub, 2024b). With its recent transition to trajectory-based submissions, the SMH provides the ideal datasets for demonstrating the utility of the energy score in a real-world context (Howerton et al., 2023; Runge et al., 2024).

Scenario projections typically aim to answer policy questions based on specific assumptions rather than predict as accurately as possible future disease trajectories. Therefore, scenario model evaluation can be complex, and must account for both the accuracy of projections against surveillance data and how well the scenario assumptions compare with reality (Howerton et al., 2023; Runge et al., 2024). This differs from short-term forecasts, where the goal is to predict as accurately as possible future disease outcomes. In practice, scenario projection evaluation often focuses on how well the projected trajectories capture the future dynamics of the epidemic (Loo et al., 2024; Reich et al., 2022; Bay et al., 2024). While other established scoring rules remain valuable and should continue to be used where appropriate, our analysis shows that the energy score is a rigorous metric for evaluating both individual models and ensemble projections, and is particularly well-suited to trajectory-based epidemic predictions. By comparing it with other scoring rules, we illustrate its distinct advantages in capturing the full dynamics of stochastic trajectories and advocate for its adoption among the tools routinely used for epidemic model evaluation.

Methods

When evaluating probabilistic predictions, it is important to use proper scoring rules for model evaluation. Proper scoring rules are evaluation measures such that a forecaster has no incentive to predict anything other than their own true belief (Bracher et al., 2021; Gneiting and Raftery, 2007). If G is the underlying generative process of the observations $\mathbf{y}$, the score comparing the observed data with G will on average give the optimal score. Let $S(P) \equiv E_{\mathbf{y} \sim G}[S(P, \mathbf{y})]$ be the expected score of a stochastic process P with respect to the generative process of the observations G (Gneiting et al., 2008). A scoring rule is strictly proper if the generative process of the observations G gives the best (minimal) expected score, and any prediction P will

produce a greater expected score unless $P = G$. This can be shown by

$$S(G) \leq S(P) \quad (1)$$

and is equal only if $P = G$. This inequality holds for negatively-oriented scores, meaning that smaller scores are considered better predictions. A score is proper but not strictly proper if the inequality holds, but is not uniquely minimized by the generative process G . This means that a prediction could give an optimal score even if it is not identical to the generative process of the observed data. The energy score is a strictly proper negatively-oriented score and it is the multivariate generalization of the continuous ranked probability score (CRPS) (Gneiting and Raftery, 2007; Gneiting et al., 2008).

The Energy Score for Trajectory-based Projections

The concept of the energy score is derived from energy statistics and it measures distances between statistical observations to quantify differences between distributions (Székely and Rizzo, 2017; Székely and Rizzo, 2013). The energy score ($ES(P, \mathbf{y})$) of a multivariate distribution P , where $\mathbf{X}^{(i)}$ and $\mathbf{X}^{(j)}$ are vectors describing realizations of random variables drawn from P , and $\mathbf{y}$ is the vector of observed values, is defined as:

$$ES(P, \mathbf{y}) = \sum_{i=1}^N p_i \|\mathbf{X}^{(i)} - \mathbf{y}\| - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N p_i p_j \|\mathbf{X}^{(i)} - \mathbf{X}^{(j)}\|, \quad (2)$$

where $\|\cdot\|$ is the Euclidean norm, p_i is the weight attributed to each individual trajectory i , and N is the total number of trajectories being analyzed (Gneiting and Raftery, 2007; Staid et al., 2017). The weights p_i can be defined such that trajectories with a higher probability of occurrence are given more weight in the evaluation (Staid et al., 2017). This approach could be applied when evaluating an ensemble model composed of multiple individual models or scenarios grouped together, where trajectories from certain models or scenarios are expected to perform better than others. In the following, we will assume that in projections from a single model, all trajectories are weighted equally, as changing the weights affects the interpretation of the energy score. If we assume that all trajectories are equally weighted with weight $p_i = \frac{1}{N}$, and we

expand the Euclidean norm, then Eq. 2 can be written as:

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$$ES(P, \mathbf{y}) = \underbrace{\frac{1}{N} \sum_{i=1}^N \sqrt{\sum_{m=1}^M (x_m^{(i)} - y_m)^2}}_{\text{distance from surveillance data}} - \underbrace{\frac{1}{2N^2} \sum_{i=1}^N \sum_{j=1}^N \sqrt{\sum_{m=1}^M (x_m^{(i)} - x_m^{(j)})^2}}_{\text{distance between trajectories}}, \quad (3)$$

where M is the number of elements in each vector (i.e. the number of time points projected), $x_m^{(i)}$ is the predicted value by trajectory i at time m , and y_m is the value of the surveillance data at time m . In this definition, the trajectory could be a time-series, where each entry is a prediction for a given date, or more generally a vector of predictions, such as different outcome targets. As we will discuss later, the score is composed of two components, the first term which compares the distance between the predicted trajectories and the observed data, and the second reflects the distances of the trajectories to each other.

The energy score in Eq. 3 is an absolute measure, meaning it is strongly influenced by deviations from the signal at higher magnitudes. In other words, if a trajectory has the same relative error for two observed values of different magnitudes, the energy score will place significantly greater weight on the deviation associated with the larger magnitude. This can be a desirable feature in scenarios where errors on large data points have more serious implications. For instance, a 30% relative error in hospitalization at the onset of an epidemic might correspond to only a few weekly admissions, while the same error at the peak of the season could represent a significant underestimation of hospital bed demand. In such cases, the energy score appropriately penalizes deviations at the epidemic peak more than those at the tail (Bracher et al., 2021; Cramer et al., 2022b).

However, in other situations, this feature may be less desirable—for example, when comparing forecast quality across states with inherently different epidemic curve magnitudes due to varying population sizes. In such cases, a relative energy score can be defined by normalizing the score by the sum of the observed time series data, thereby rescaling the score by the signal’s overall size (Gneiting et al., 2008). A normalized energy score can be written as:

$$ES_{\text{norm}}(P, \mathbf{y}) = \frac{ES(P, \mathbf{y})}{\sum_{m=1}^M y_m}, \quad (4)$$

which provides a relative measure that facilitates comparisons of normalized energy scores across different locations. This normalization still emphasizes deviations at the peak within each state but adjusts the overall score based on the sum of the signal, enabling fairer comparisons between states with different

population and outbreak sizes (Cramer et al., 2022b). This approach can also be extended to compare scores across different projection targets or time periods. Other normalization methods, typically applied to the WIS, have been proposed for epidemic model evaluation. These include a logarithmic transformation of scales (Bosse et al., 2023), rescaling by the standard deviation across models (Howerton et al., 2023), and pairwise model comparison using a geometric mean (Cramer et al., 2022b). While these methods can be applied to the energy score, we choose to normalize by the surveillance data magnitude as this approach maintains importance of the epidemic peak and avoids relying on other model projections. The issue of normalization is similarly relevant in the multi-dimensional extension of the energy score, as discussed in the next section. Unless otherwise specified, we will use the energy score as defined in Eq. 3.

Multi-Dimensional Energy Score

One benefit of the energy score is that it can be adapted into a performance measure across multiple dimensions. This would allow us to evaluate a model across multiple target outcomes (i.e. cases, deaths, or hospitalizations), age groups, locations, and so on, with a single score, giving us a comprehensive understanding of a model's performance with respect to all of its predictions in high-dimensional space. With other scoring rules, this would only be possible via a summary statistic, such as a sum or average of scores for each outcome variable. To calculate this, we look at each time point as a multi-dimensional vector, where T represents the number of prediction targets we are assessing. Now, we have a matrix where the columns describe the predictions at each time point, and the rows show the time-series predictions for each outcome. In this definition of the energy score, we are looking at the distance between matrices instead of the distance between vectors, as in the standard energy score. Therefore, we can use the Frobenius norm to measure distance, as it is the multi-dimensional extension of the Euclidean norm, and define the multi-dimensional energy score as:

$$\text{ES}_{\text{dim}}(A, Y) = \frac{1}{N} \sum_{i=1}^N \sqrt{\sum_{j=1}^T \Phi_j^2 \sum_{m=1}^M (a_{jm}^{(i)} - y_{jm})^2} - \frac{1}{2N^2} \sum_{i=1}^N \sum_{k=1}^N \sqrt{\sum_{j=1}^T \Phi_j^2 \sum_{m=1}^M (a_{jm}^{(i)} - a_{jm}^{(k)})^2}, \quad (5)$$

where $A^{(i)}$ is a $T \times M$ matrix of predictions for trajectory i , with T outcomes (i.e. prediction dimensions) and M time points, Φ_j is a normalization factor for the magnitude of the signal along each dimension/target, and N is the number of trajectories reported for each outcome target. The $T \times M$ matrix Y describes the surveillance data, matching the construction of the trajectory matrices $A^{(i)}$.

The factor Φ_j^2 rescales the signal, adjusting the contribution of each forecast target T to the multi-dimensional energy score. When $\Phi_j^2 = 1$, targets with larger magnitudes dominate the score, implying that they are more relevant for assessing model performance. However, this assumption may not always be desirable. For example, if the targets are hospitalizations and deaths, the $T = 2$ energy score will be heavily influenced by the hospitalization target, which typically has a much larger magnitude than the death target. However, it may be preferable for the model to predict both targets with equal accuracy. In such cases, we can use a rescaling factor $\Phi_j = \frac{1}{\sum_{m=1}^M y_{jm}}$, dividing by the sum of the observation vectors for each target outcome to ensure that all outcome dimensions contribute similarly to the multi-dimensional energy score. The same principle applies when forecast targets correspond to different geographical locations. For instance, if each U.S. state is treated as a forecast target, $T = 50$, then the energy score will be dominated by states with larger populations, which typically have higher hospitalization or death counts. Applying a rescaling factor in this context ensures that performance across all states is weighted equally, regardless of population size. The choice of Φ_j^2 can be adapted on a case-by-case basis, depending on the objectives of the energy score assessment and the desired balance between different forecast targets.

Finally, when constructing the multi-dimensional trajectory matrices, trajectories must be paired appropriately across dimensions. This means that trajectories of different targets (T) from the same simulation (i) must be combined into the same $T \times M$ matrix $A^{(i)}$ to preserve correlations between dimensions, as the construction of the time series matrix directly influences the energy score value. There can be cases where the trajectories across dimension do not originate from the same simulation trajectory. When trajectories for each target are generated independently across dimensions, any pairing is acceptable since there are no cross-dimensional correlations. We discuss these cases further in the Results and Supplementary Material. Regardless of the generation method, all matrices must have the same dimensions (i.e. there must be the same number of trajectories for each target). For example, with 100 trajectories in dimension 1 and 200 in dimension 2, only 100 trajectory matrices can be constructed.

Results

Comparison of proper vs. strictly proper scoring rules

Several scoring rules have been proposed in the literature to evaluate the performance of probabilistic epidemic projections. Among these, the weighted interval score (WIS) has emerged as a widely adopted

standard in forecasting and scenario modeling efforts (Bracher et al., 2021; Howerton et al., 2023). The WIS is a negatively-oriented proper score applied to prediction intervals given by a model’s quantiles. The score consists of three terms that describe the width or uncertainty in the prediction interval, and a penalty if the surveillance data lies outside the prediction interval. It is computed at each time point with prediction P and observed value y as a weighted sum of the interval score for each prediction interval of interest, and approximates the CRPS (Bracher et al., 2021). In order to evaluate the performance of a full projection time series using the WIS, we take the average of the WIS calculated at each time point, as this ensures that the score remains proper (Bracher et al., 2021). We further discuss the WIS and CRPS in the Supplementary Material.

The energy score and WIS both evaluate a projection based on its calibration, or the distance between the predicted and observed values, and sharpness, which is the amount of uncertainty given by the prediction. We show in Fig. S1A that the energy score and WIS are similar when evaluating a synthetic predictive distribution at a single time point. However, there are differences between the two metrics that must be considered. Most importantly, the energy score is strictly proper for the full projection time series where the WIS is proper, but not strictly proper. This has implications for how these scores evaluate specific probabilistic predictions. We present proofs for the propriety of the energy score and WIS in the Supplementary Material.

To provide a visual intuition of this difference, we generate trajectories from two stochastic epidemic processes that have the same marginal distribution at each time step. First, we generate trajectories from a stochastic SIR model using a chain binomial process. We parameterize the SIR model such that the transmissibility is $\beta = 0.625$ and the recovery rate is $\mu = 0.25$ giving us a reproduction number $R_0 = 2.5$. For the second process, we create a noisy SIR model, where we randomly shuffle the values for the number of infectious individuals at each time point in the SIR model trajectories. Then we randomly group these values across time to produce noisy epidemic trajectories. This gives us two epidemic-like processes with the same marginal distribution at each time step, one with time-correlated, and one with time-uncorrelated stochastic trajectories. An example of the two trajectory processes are shown in Fig. 1A and 1B, where we also show the quantile format for these trajectories in Fig. 1C, which is identical for each trajectory process. We then score the two processes using as observation vector a single realization of the stochastic SIR model described above. Using this framework, we calculate the energy score and WIS for 200 iterations, generating 100 trajectories at 60 time points for each model at every iteration.

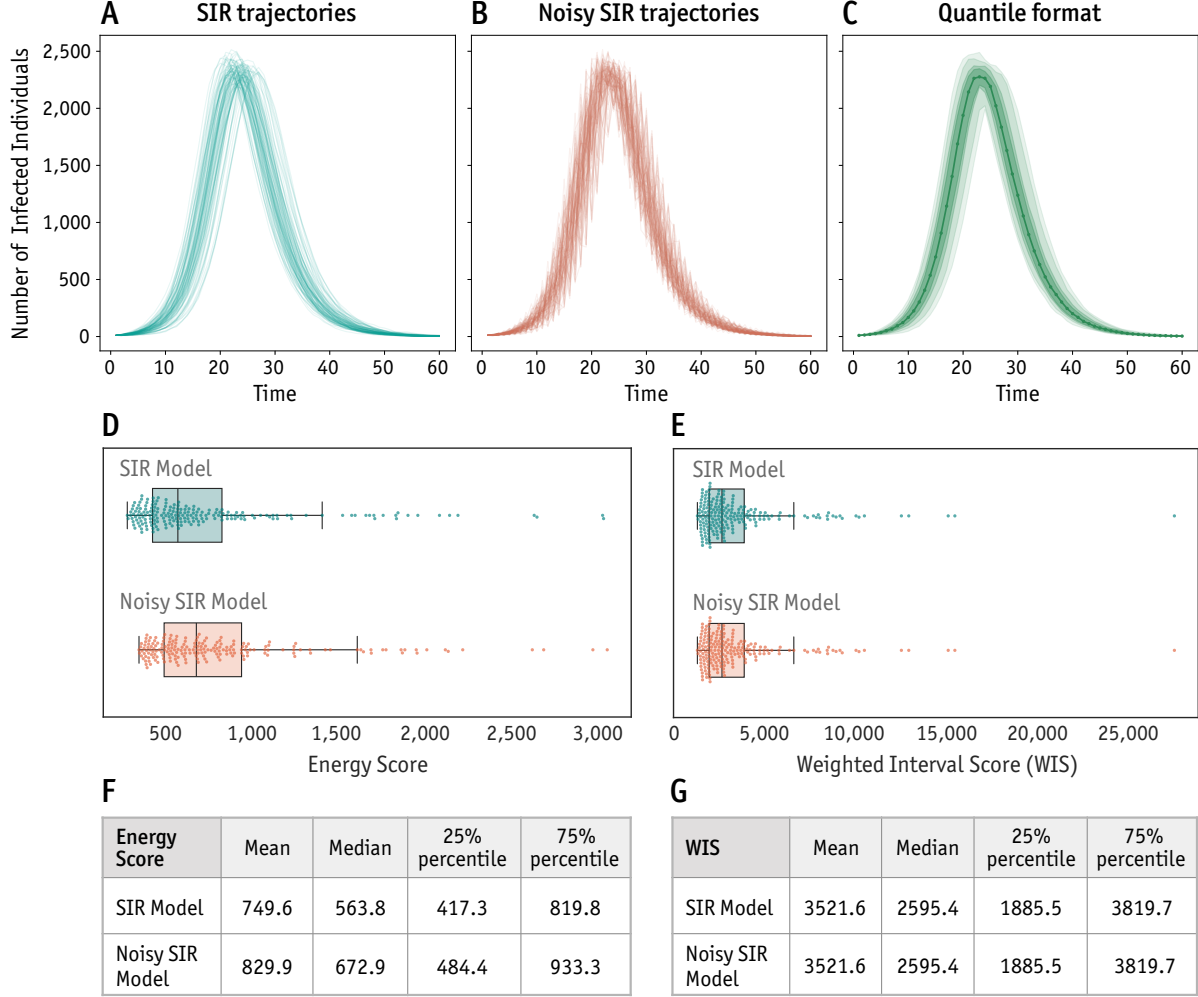


Figure 1: Synthetic experiments comparing energy score and WIS propriety and evaluation. Comparison of the WIS and energy score for (A) SIR model and (B) noisy SIR epidemic trajectory process with the same marginal distribution at each time point, generating the same quantile format (C) with median (solid line) and 50%, 70%, 90%, and 98% prediction intervals shaded. Boxplot of the energy score (D) and WIS (E) obtained from each trajectory process given 100 trajectories each, showing the distribution for each score and epidemic trajectory process across 200 iterations. For visualization purposes, the maximum values for each score and model type is not shown. Boxplots are created such that the box shows the 25%, 50% and 75% quantiles, and the whiskers represent $1.5 \times$ interquartile range (IQR). Descriptive statistics of the (F) energy score and (G) WIS for the SIR and noisy SIR model.

In Fig. 1D and E, we show the distribution of the scores for each epidemic trajectory process, and descriptive statistics of these distributions in Fig. 1F and G. From the boxplots and tables of descriptive statistics, we find that the WIS for both model processes is the same, but the energy scores differ. The energy score is able to distinguish between two processes with the same marginal distribution but differing individual behavior, where the WIS scores them identically. The SIR model process, on average, produced better values of the energy score than the noisy model, which agrees with the knowledge that the observed

values were generated from the SIR model. This difference is due to the energy score being strictly proper, while the WIS is proper but not strictly proper. If a prediction P has the same marginal distribution as the true underlying process G , it would give the ideal WIS score to both even if $P \neq G$; on the other hand since the energy score is strictly proper, only a prediction $P = G$ can give the ideal energy score (Gneiting and Raftery, 2007). Moreover, the fact that on average the energy score gives lower scores to the SIR model demonstrates the impact that the additional information contained within individual trajectories provides. Since the noisy SIR model trajectories do not follow the same temporally correlated behavior as the SIR model, and therefore the observation vector, they are penalized.

Assessing Forecast Models with Multiple Peak Timing: Advantages of the Energy Score

A fundamental distinction between the energy score and the WIS lies in how they capture the epidemic dynamics. While the energy score evaluates the complete set of individual trajectories directly, the WIS relies on summary statistics, typically quantiles, computed at each fixed time point. This difference has important implications for how well each scoring rule captures key dynamic features of epidemic forecasts, such as the timing and magnitude of epidemic peaks.

Fixed-time quantile summaries, as used in the WIS, tend to obscure multi-modal or non-synchronous behaviors across trajectories. For example, consider a model in which each trajectory includes a single epidemic peak, but the peak time of the trajectories vary. When trajectories are aggregated into quantiles at each time point, the resulting summary can flatten or even mask these peaks entirely. The quantile representation may suggest a smoother, less peaked epidemic curve that does not correspond to any individual realization of the model (Juul et al., 2021).

This effect is especially problematic in the context of scenario modeling. With scenarios, models should not only predict the most likely outcome, but also explore plausible alternative trajectories, including worst-case outcomes. When such forecasts are scored using the WIS, models that correctly capture complex behaviors, such as epidemics having a wide distribution of peak time, may be unfairly penalized if their median or central quantile predictions deviate from the observed data, even if a substantial subset of trajectories match the observed epidemic.

To illustrate this limitation, we designed a synthetic forecasting experiment involving the superposition of two stochastic SIR models, each producing a single epidemic peak at a different time. We generate a standard stochastic SIR model (as described in the previous section) with low variability in the peak timing.

Additionally, we generate trajectories from a double-peak SIR model with trajectories having a different peak distribution: each realization contains a single peak, but across the ensemble, peaks occur at two distinct time points. Fig. 2A shows the trajectories and corresponding quantile representations of the single peak and double-peak SIR models. For evaluation, we selected a single realization of the single peak SIR model as the ground truth, and we compare forecast scores using both the energy score and the WIS. For the WIS calculation, quantiles were estimated from the trajectories generated by the SIR models. We repeat this process generating 100 trajectories and scoring each model for 500 iterations, and test different widths between the two peaks in the trajectories, namely peaks that are separated by 5, 10, or 20 time steps.

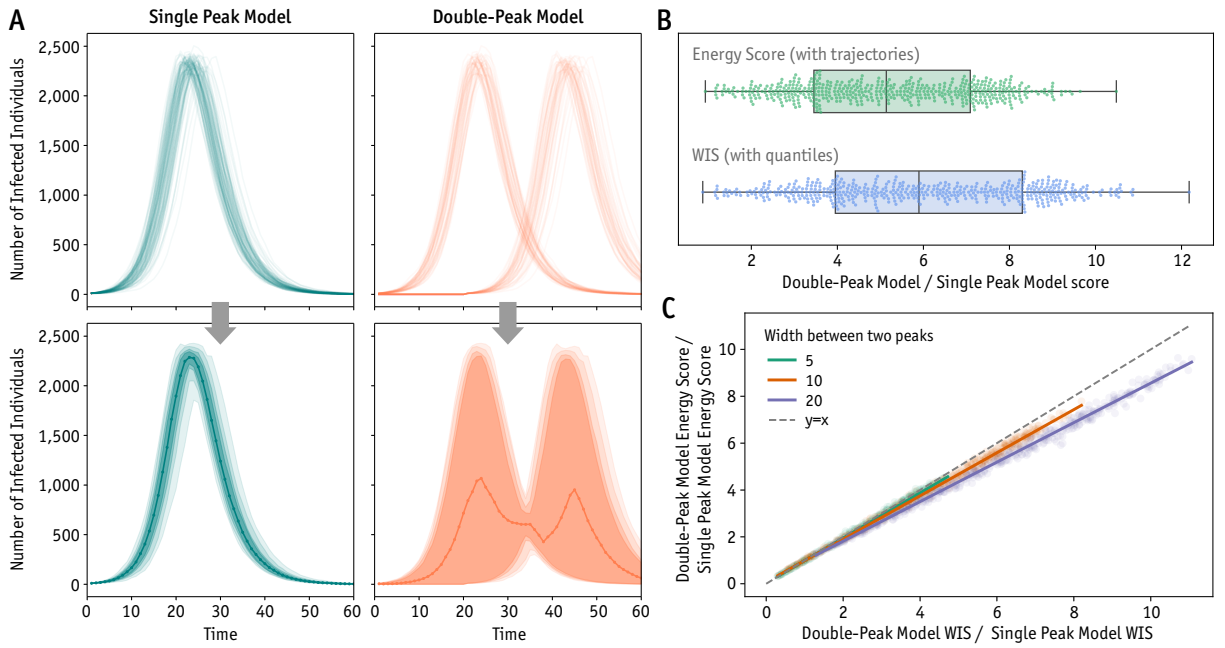


Figure 2: (A) Single peak (left) and double-peak (right) stochastic SIR model trajectories (top) and corresponding quantile format for the 50% and 98% prediction interval (bottom). Shown for a width between the two peaks of the trajectories of 20 time steps. (B) Distribution of the ratio between double-peak model and single peak model score for the energy score and WIS for 500 iterations of generating 100 trajectories for each model, with a distance of 20 time steps between the 2 peaks in the double-peak model's trajectories. Boxplots are such that the box shows the 25%, 50% and 75% quantiles, and the whiskers represent 1.5 $\times$ interquartile range (IQR). (C) Relationship between the double-peak model / single peak model score ratio for the energy score and WIS for different distances between the two peaks in the model. Dashed gray line shows $y=x$ and the solid lines show the linear regression fits of the data for the different widths between peaks. Dots show the scores for each iteration of trajectory generation and scoring.

We compare the behavior of the two scores by finding the ratio between each score given by the double-peak model divided by the corresponding score for the single peak model. As shown in Fig. 2B, the forecast model with the double-peak structure is significantly more penalized by the WIS than by the energy score. The ratio between the double-peak model and single peak model are typically lower for the energy score than the WIS. This discrepancy increases with the growing separation between the two peaks, as confirmed

by our correlation analysis of the relative scores in Fig. 2C. As the two peaks grow farther apart, the ratios
for the energy score and WIS become increasingly different. The energy score, which directly evaluates
the distribution of complete trajectories, is able to identify subsets of realizations that accurately match the
observed dynamics.

These findings indicate a potential limitation of quantile-based evaluations in capturing complex epi-
demic behaviors. When model outputs display substantial peak-timing uncertainty or multi-modal epidemic
dynamics, the energy score offers a more appropriate measure of predictive performance, as it directly eval-
uates the full set of stochastic trajectories. This property makes it particularly well-suited for assessing
forecasts and projections reported in trajectory format. Figure S2 illustrates how similarly complex dynam-
ics can lead to marked differences in scoring during real-time epidemic projection exercises.

Case Study from the Flu Scenario Modeling Hub

We illustrate the application of the energy score to epidemic scenario projections for the 2023-24 pro-
jection round of the Flu Scenario Modeling Hub. In previous projection rounds, modeling teams were only
required to report quantiles of their predictions aggregated for each week, but beginning in the 2023-24 in-
fluenza round, teams submitted at least 100 individual trajectories from their model output for each scenario,
and location (Loo et al., 2024). The objective of this projection round was to explore the implications of
different vaccine coverage levels (high, normal, or low) and the dominant circulating strain (either A/H3N2
or A/H1N1) on the trajectory of weekly hospitalizations during the 2023-2024 flu season for US states and
nationally (Scenario Modeling Hub, 2024a). Modeling teams reported projections for these 6 scenarios,
from September 3, 2023 to June 1, 2024. Further information about the projection round and model output
can be found in the Flu SMH’s [GitHub repository](#). Table S1 describes the 2023-24 SMH scenarios in further
detail. In Fig. 3, we show the reported model output for the ‘MOBS_NEU-GLEAM_FLU’ model in the US
nationally for all 6 scenarios of the 2023-24 Flu SMH round. We see how the scenarios result in distinct
epidemic outcomes given assumptions on the circulating influenza strain and vaccine uptake. We analyze
the incident hospitalization projections from September 9, 2023 to April 27, 2024.

Given the widespread use of the weighted interval score (WIS) for evaluating probabilistic epidemic
predictions, we compare the performance assessment provided by the WIS to that of the energy score to
understand how these metrics evaluate model projections. We further describe how the WIS is computed
from the Flu SMH projections in the Supplementary Material.

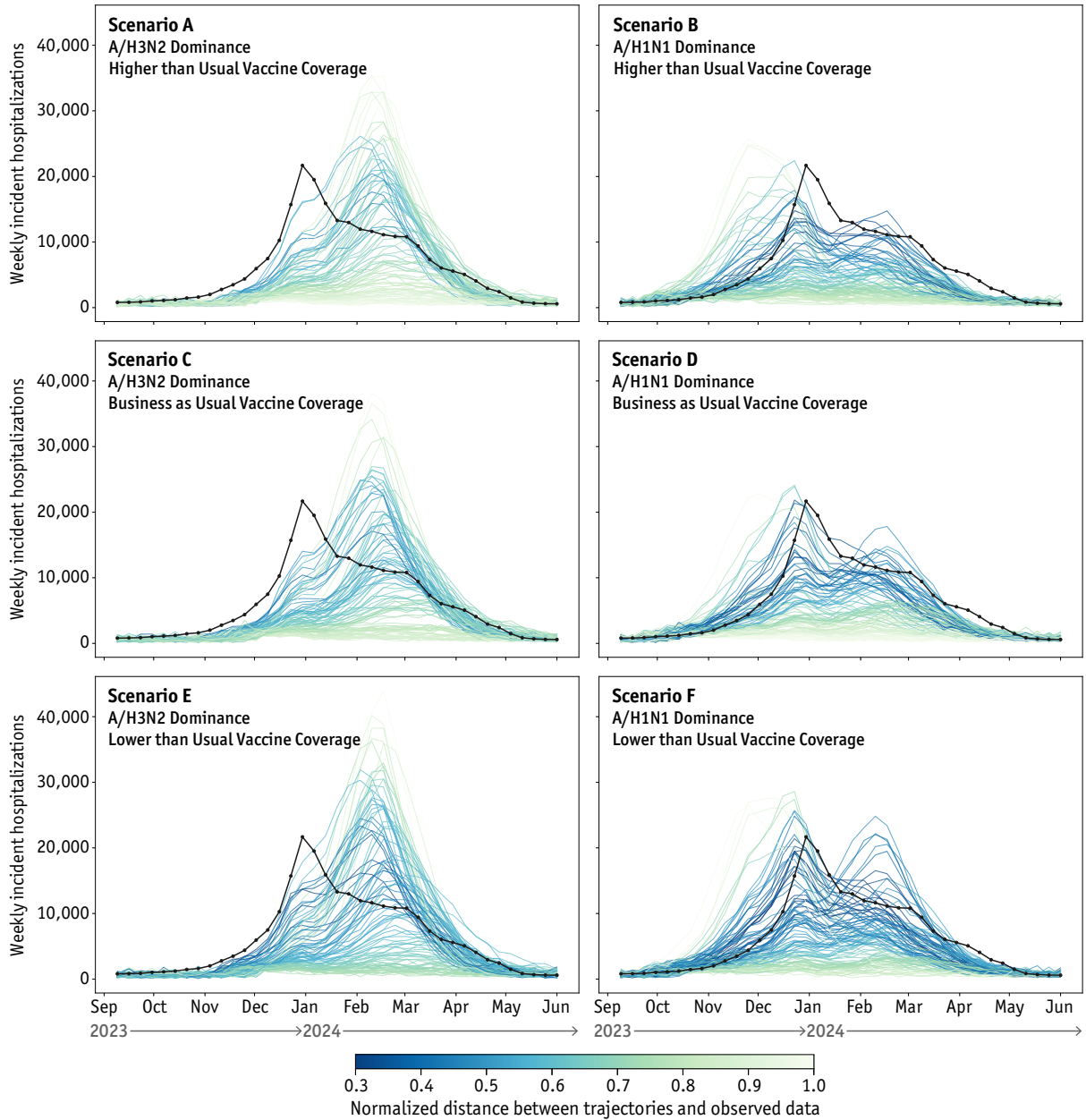


Figure 3: Epidemic predictions in the trajectory format. One hundred trajectories for incident hospitalizations for the ‘MOBS_NEU-GLEAM_FLU’ model for all 6 scenarios in the Flu Scenario Modeling Hub 2023-24 round 1 with observed surveillance data (black) nationally in the United States. Trajectories are colored by their normalized distance from the observed data. Lighter colors represent a larger distance.

Single Model Evaluation

We focus the first part of our analysis on the ‘MOBS_NEU-GLEAM_FLU’ model submitted to the Flu Scenario Modeling Hub, which is a multi-scale, age-structured, stochastic metapopulation epidemic model that uses global flight and commuting data to simulate the spread of an infectious disease (Balcan et al.,

2010; Chinazzi et al., 2024). In this section, we examine the energy score for this single model across time periods, locations, and scenarios.

As noted earlier, the energy score is influenced by the magnitude of values defined in each trajectory. This dependency complicates the comparison of scores across locations, time periods, or target outcomes with differing magnitudes. In Fig. 4A, we compare the energy score for the ‘MOBS_NEU-GLEAM_FLU’ model influenza hospitalization projections for each scenario and location to the sum of the corresponding surveillance time series. We find that the non-normalized energy score is strongly correlated to the size of the signal. However, Fig. 4B shows that the normalized energy score depends much less on the magnitude of the surveillance data, with an R^2 of 0.018. This means that scores can be compared with the normalized energy score, and demonstrates its nature as a relative score.

While the energy score is defined for a multivariate time series, we analyze its behavior at individual time points to study how the energy score changes throughout an outbreak. In Fig. 4C, we show the energy score at each week of the flu season (Eq. 3 where $M=1$) for the ‘MOBS_NEU-GLEAM_FLU’ model, scoring incident hospitalization projections for Scenario D (A/H1N1 Dominance, Business as usual vaccination coverage) nationally in the U.S., and we decompose these scores to look at the contribution of individual terms. The energy score is made up of a positive term describing the distance between the trajectories and surveillance data, and the pairwise distance between all trajectories subtracted from this value. This negative term is why the energy score value is less than the component describing the distance from surveillance data. In this example, the energy score is largely composed of the distance between each trajectory and the observed data. The score increases at time points near the peak as expected because the energy score is an absolute metric depending on the magnitude of the surveillance signal, which means it will typically give higher weights near the peak of an epidemic curve. We show in Fig. S1B that the WIS follows similar patterns when evaluated at each week of the influenza season.

Since the WIS is a commonly used score for evaluating epidemic projections, we compare the performance of the ‘MOBS_NEU-GLEAM_FLU’ model using the energy score and WIS. If we calculate the energy score and WIS for the predictions made for each location and scenario for incident hospitalizations in the 2023-24 Flu Scenario Modeling Hub round, we find a strong correlation between the two scores, shown in Fig. 4D. This highlights that a prediction that performs well under the WIS is likely to also be scored well by the energy score. In Fig. S1C, we show a strong correlation between the normalized energy score and normalized WIS as well.

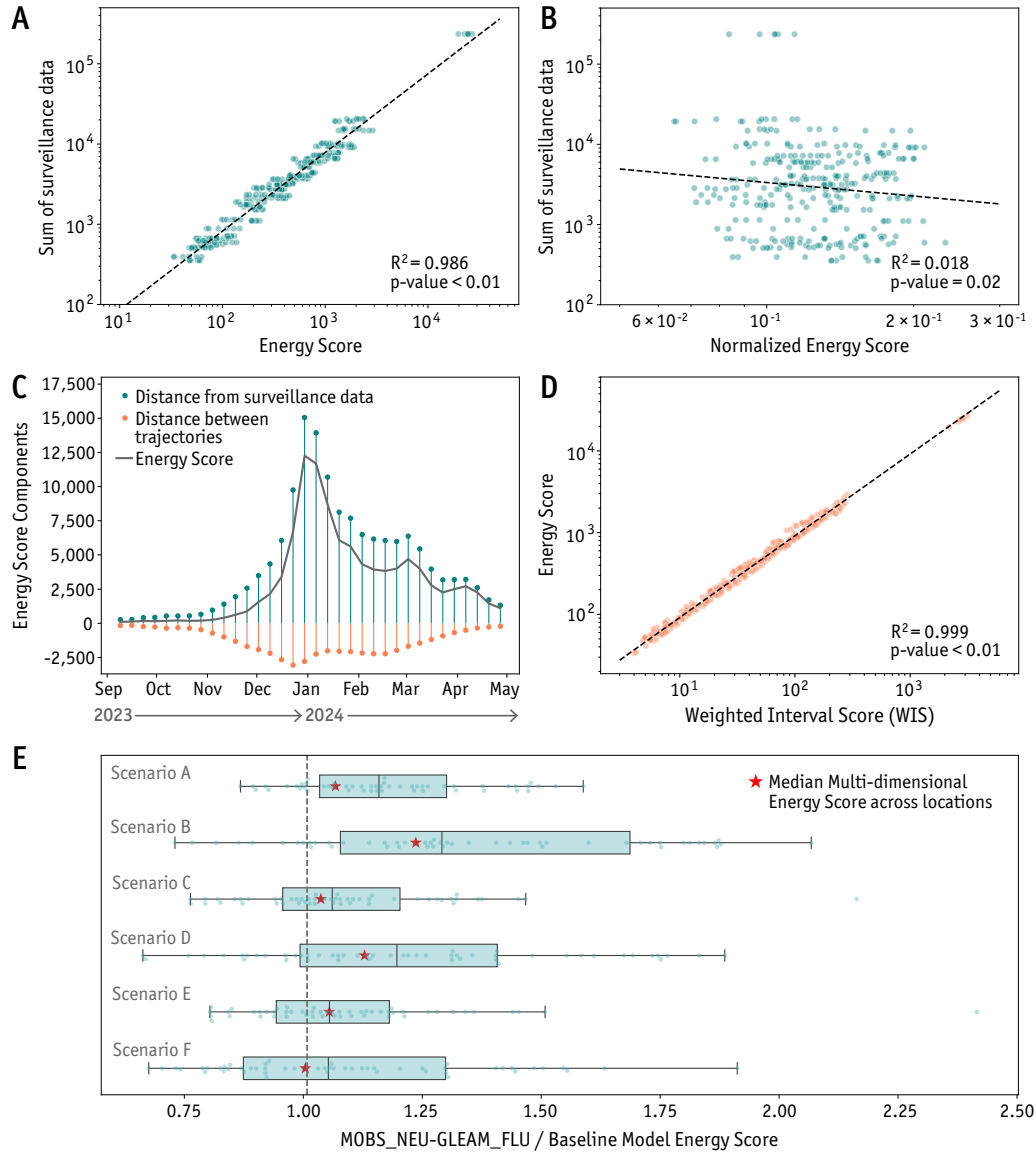


Figure 4: Evaluation of a single model with the energy score. Comparison of the dependence of the (A) energy score and (B) normalized energy score on the sum of the surveillance data of each scenario and location for the model projections. (C) Decomposition of the energy score terms at each week for the Scenario D projections, where the blue represents the term describing the distance of the trajectories from observed data, and the orange describes the term representing the distance between all pairs of trajectories. The gray line shows the full energy score calculated at each week. (D) Relationship between the WIS and energy score. Each point describes the WIS and energy score for a location and scenario. The WIS was found by estimating the quantiles estimated from the trajectories. (E) Boxplot of energy score ratio across 52 locations for the ‘MOBS_NEU-GLEAM_FLU’ model compared to a 4-week-ahead naive baseline model for each scenario. Vertical dashed line shows where the ‘MOBS_NEU-GLEAM_FLU’ and baseline model have the same scores, where ratios below one describe when the model performs better than the baseline. The overlaid red stars show the median of the multi-dimensional energy score ratio across locations for 50 iterations of randomizing the trajectory pairings. Boxplots are created such that the box shows the 25%, 50% and 75% quantiles, and the whiskers represent $1.5 \times \text{IQR}$. Scatter plots (A,B,D) show a dashed line of the linear regression fit, with the R^2 and two-sided p-value describing the fit. All results are shown for the ‘MOBS_NEU-GLEAM_FLU’ model 2023-24 Flu SMH incident hospitalization projections.

Using the energy score, we are also able to compare performance across scenarios and locations. In Fig. 4E, we evaluate the performance for all scenarios, looking at the distribution of scores for predictions at each location. To evaluate model performance, we create a naive baseline forecast for 4-week-ahead projections, to use as a reference point against which we compare the performance of long-term scenario projections (Ray et al., 2023). While alternative baseline methods can be devised for scenarios projections, this approach has been used previously to assess the performance of several rounds of the SMH COVID-19 projections (Howerton et al., 2023). The baseline forecast generates projections where the median is equal to the most recently observed data point, with expanding uncertainty for longer horizons informed by historical changes in incidence counts for consecutive weeks. We report additional details on the generation of the naive baseline model and its trajectories in Fig. S3 in the Supplementary Material. From the baseline model, we create an energy score ratio where we divide the energy score for each model, location, and scenario by the energy score of the baseline model for each location. A ratio less than 1 describes when the scenario model performs better than the naive baseline model. It is important to note that scenario projections are not necessarily expected to outperform the baseline model. This is because the baseline model is continuously updated with new data each week to produce 4-week-ahead predictions, whereas scenario projections are made several months in advance without incorporating recent surveillance data. In our analysis, we find that most scenarios for the ‘MOBS_NEU-GLEAM_FLU’ model perform similar to or slightly worse than the naive baseline model. In Table S2, we show descriptive statistics for the energy score ratio ‘MOBS_NEU-GLEAM_FLU’ model for each scenario.

We also utilize the multi-dimensional energy score to compute a single comprehensive score for each scenario, providing an overall assessment of the ‘MOBS_NEU-GLEAM_FLU’ model’s performance across all U.S. states. The multi-dimensional energy score requires trajectories to be paired consistently across dimensions, including across states. However, the SMH does not explicitly provide the pairing of trajectories across different locations. For the ‘MOBS_NEU-GLEAM_FLU’ model, trajectories in different states are originated from independent calibrations, allowing us to randomize the trajectory identifiers and conduct a sensitivity analysis to evaluate how different pairings influence the multi-dimensional energy score, repeating this process over 50 iterations. Additionally, we calculate an energy score ratio, as previously described, to compare these values against the multi-dimensional energy score of the 4-week-ahead naive baseline model. In Fig. 4E, the red stars show the median multi-dimensional energy score ratio across these iterations for each scenario, where the dimensions are U.S. states. In Table S5, we show that the un-

certainty around the multi-dimensional energy score for randomizing the trajectory pairings is very small. 342
 Additionally, we find that the distribution of the energy score ratio across locations in the boxplots and the 343
 multi-dimensional energy score rank the scenarios similarly, with both medians following the same pattern. 344
 This shows that the multi-dimensional energy score is evaluating the model similarly to the standard energy 345
 score, but in an aggregated manner. In addition, we examine the impact of removing the normalization fac- 346
 tor from the multi-dimensional energy score expression in order to keep emphasis on locations with larger 347
 signals in Fig. S4. 348

Multi-Model Analysis 349

In this section, we focus our analysis on the performance of all models submitted to this Flu SMH 350
 projection round that provided incident hospitalization projections for multiple locations. This selection 351
 resulted in six individual models being included in the analysis, with four models excluded from the study 352
 because they only reported data for one location. Ensemble models generated by the SMH were not con- 353
 sidered, as these models report results only in a quantile format. To illustrate the utility of the energy score 354
 in comparing performance across multiple models, we analyze model rankings and performance across dif- 355
 ferent locations. In Fig. 5 we show how the energy score varies across models, scenarios, and locations. 356
 We calculate the energy score ratio in comparison with the 4-week-ahead naive baseline model (Ray et al., 357
 2023) described previously for each scenario, location, and model, where a ratio below one represents a case 358
 where the scenario model performed better than the naive baseline model. We find that there are models 359
 that consistently perform better than others, and that some models perform very similar to or slightly better 360
 than the baseline model. Table S3 presents descriptive statistics of this data for all scenarios and models by 361
 summarizing the energy score ratio for all models and scenarios examined. This demonstrates the ability of 362
 the energy score to compare models and differentiate between better-performing models. This tells us about 363
 relative model performance within each scenario. 364

We also investigate how model rankings based on the energy score compare to those derived from the 365
 WIS. For each scenario and location, we rank the models from lowest to highest score using both the energy 366
 score and WIS. To assess the agreement between these rankings, we calculate Kendall's τ rank correlation 367
 across all models. The Kendall's τ correlation coefficient compares rankings by counting the number of 368
 pairs of objects that are in an incorrect order, divided by the total possible pairs (Kendall, 1938). A rank 369
 correlation of 1 means that the rankings of the energy score and WIS are identical. In Fig. S1D, we show a 370

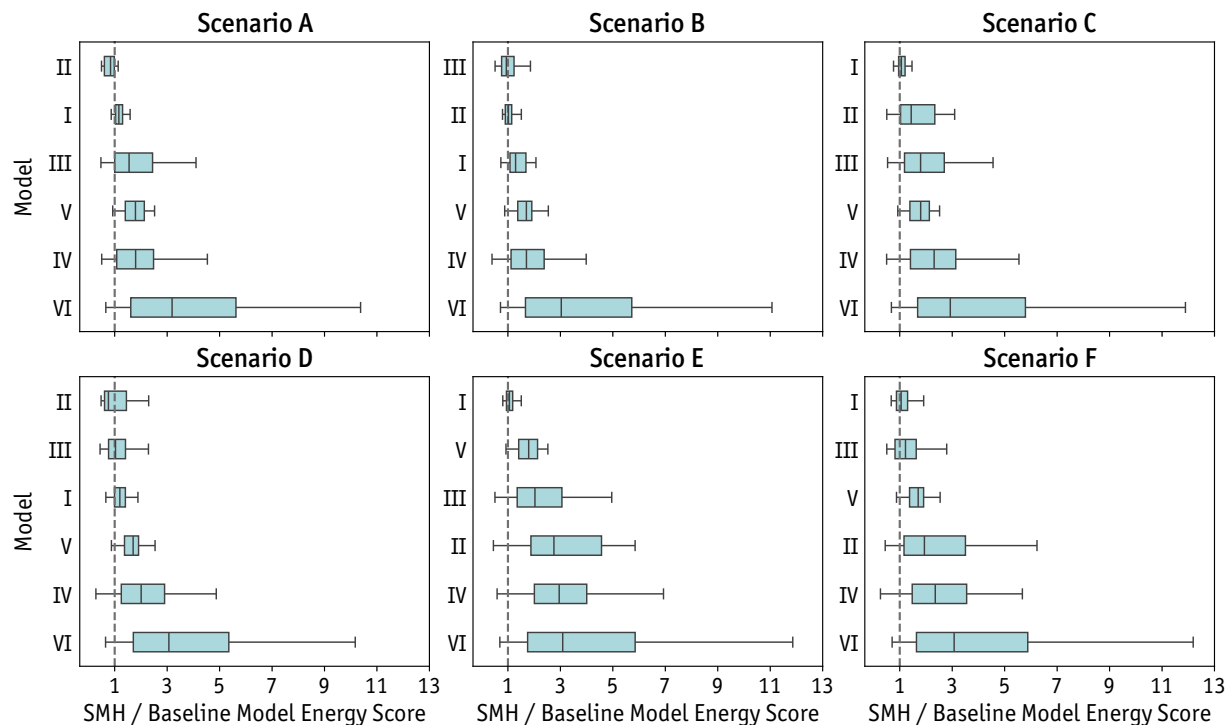


Figure 5: Energy score ranking across models. Boxplots showing the distribution of the ratio of the energy score for each model, location, and scenario in Flu Scenario Modeling Hub divided by the energy score for the 4 week ahead naive baseline flu model at each location. Each boxplot shows the energy score ratio distribution over the number of locations the corresponding model reported (shown in Table S3). Vertical dashed line shows where the ensemble and baseline model has equal energy scores such that ratios below one describe when the SMH model performs better than the baseline. Models are ordered by median energy score within each scenario. Boxplots are such that the box shows the 25%, 50% and 75% quantiles, and the whiskers represent 1.5 $\times$ IQR. Note that Model I represents the ‘MOBS_NEU-GLEAM_FLU’ model, which can be compared to results in Fig. 4.

histogram of the Kendall’s τ rank correlation between the energy score and WIS for all models and scenarios. 371
The mean rank correlation coefficient between the energy score and WIS is 0.87. More precisely, we find 372
that for 49.7% of projections, the energy score and WIS have a Kendall’s τ correlation coefficient of 1. This 373
shows that the energy score and WIS rank models quite similarly, but in a majority of cases, the rankings 374
are not identical. 375

Trajectory-Based Ensemble 376

In multi-model analysis, it is common practice to create an ensemble model by combining predictions 377
from the multiple individual models. (Ray et al., 2023; Reich et al., 2022). In the ensemble, the information 378
and uncertainty from multiple models are aggregated together to provide a “consensus” projection of future 379
possibilities aggregating the different assumptions and methodologies of individual models (Cramer et al., 380
2022b). The performance of ensemble models has been shown to generate improved future predictions 381
that characterize uncertainty better than individual models (Bates and Granger, 1969; Cramer et al., 2022b; 382

McGowan et al., 2019; Reich et al., 2022). The SMH includes three ensemble models as part of its analysis. 383
However, these ensembles are built by first converting each model’s trajectories into a quantile format before 384
applying the ensemble methods. Here, we propose an alternative approach for generating an ensemble 385
model that directly utilizes the trajectories reported by each modeling team. In this method, we simply 386
bundle all trajectories from each model into the definition of a single ensemble, assigning equal weight 387
to each trajectory. In our case, each model contributes the same number of trajectories. If this is not the 388
case, weighting or sampling methods could be considered so that models reporting more trajectories do not 389
have an disproportionate influence on the ensemble behavior. This ensemble approach avoids the need to 390
summarize stochastic model outputs into quantile format, leveraging the raw data provided by the models 391
instead. In Fig. S5, we compare this ensemble of trajectories method with the three ensemble models 392
reported by the SMH using the WIS. We cannot calculate the energy score of the SMH-reported ensemble 393
models because they are only reported in quantile format, so trajectories are not available. 394

In Fig. 6, we explore the energy score across locations using the trajectory-based ensemble. We include 395
all models reported to the 2023-24 Flu SMH that submit projections of incident hospitalizations for any 396
number of the 50 U.S. states, Washington D.C., or nationally. Fig. 6A shows the ratio of the energy score of 397
the trajectory-based ensemble model divided by the energy score of the 4-week-ahead naive baseline model 398
at each location (U.S. states and the District of Columbia) for Scenario D. Locations in blue show where the 399
ensemble model performed better than the baseline model (ratio < 1), where those in orange show where 400
the naive baseline model had better performance (ratio > 1). We observe heterogeneous performance across 401
the United States, with the ensemble model outperforming the naive baseline in 24 locations, but falling 402
short in 27. These results indicate areas where model performance could be improved for future predictions. 403
Notably, in this example, the ensemble tends to perform better in the Southern U.S. states. 404

We extend this to show the distribution in the energy score ratio compared to the 4-week-ahead naive 405
baseline model across locations for each scenario in Fig. 6B. This allows us to assess the overall performance 406
of the trajectory-based ensemble model for each scenario, and compare the scenarios to each other. This 407
analysis illustrates that scenarios B, D, and F are the best-performing scenarios for the ensemble model, 408
which corresponds to the scenarios describing H1N1 as the dominant circulating strain. This is in agreement 409
with the observed strain dynamics in the United States for the 2023-24 influenza season (Centers for Disease 410
Control and Prevention, 2024). We show summary statistics of this data in Table S4. 411

It is worth noting that we do not calculate the multi-dimensional energy score for the ensemble model 412

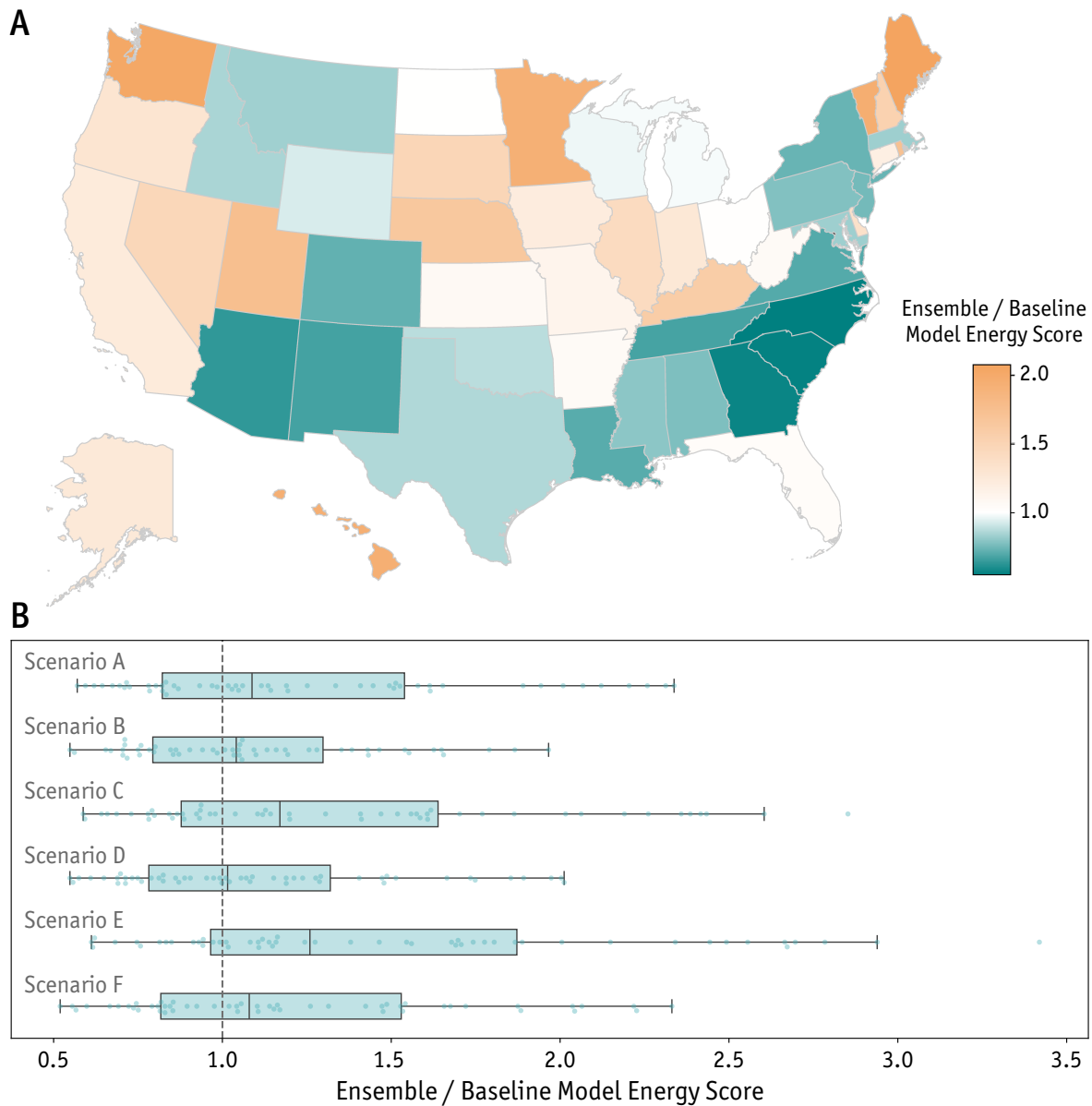


Figure 6: Energy score for the trajectory-based ensemble model. (A) Map of the US showing the energy score ratio (trajectory-based ensemble / baseline model energy score) for each US state of the 2023-24 Flu Scenario Modeling Hub round for Scenario D. Ratios below one (blue) show where the energy score for the trajectory-based ensemble model for a state was better than the naive baseline, where the ratios above one (orange) show where the baseline model had better performance. (B) Boxplots of the energy score ratio across 52 locations for all scenarios compared to a 4-week-ahead naive baseline model. Vertical dashed line shows where the ensemble and baseline model has equal performance, where ratios below one describe when the ensemble model performs better than the baseline. Boxplots are such that the box shows the 25%, 50% and 75% quantiles, and the whiskers represent $1.5 \times \text{IQR}$.

due to the assumptions required about trajectory independence and pairing—information that is unavailable 413
for the other modeling teams within the SMH. The multi-dimensional energy score relies on constructing 414
matrices that group trajectories across dimensions, so unknown dependencies could distort the score by 415

introducing unexpected dependencies. Ignoring these assumptions could bias the multi-dimensional energy score and compromise the results of these calculations.

This ensemble of trajectories method could be easily translated to a different weighting process where models or even specific trajectories are weighted differently as data is gathered (Sherratt et al., 2024). This allows for a flexible framework for ensemble creation and performance analysis using only single trajectories generated by individual models without the need to summarize and aggregate model output using quantiles.

Discussion

The energy score is not a new metric and has been widely applied to the evaluation of probabilistic forecasts across diverse fields (Gneiting et al., 2008). Yet, its potential in epidemic modeling has remained largely underexplored. The increasing adoption of trajectory-based submissions in epidemic forecasting underscores the need for evaluation methods that can capture their full complexity. In this context, our findings show the value of the energy score for improving how forecasts are assessed, how uncertainty is communicated, and how model outputs are compared across settings.

Using multiple scoring rules to assess predictions is shown to be useful (Pic et al., 2025), and the energy score adds a strictly proper performance measure to the toolbox of probabilistic prediction analysis methods. The energy score also provides a principled framework for evaluating epidemic model outputs in trajectory format, avoiding the need of the additional aggregation step to summarize stochastic realizations into quantiles. By operating directly on individual trajectories, it captures the full temporal structure and variability of the projections, thereby preserving information that would be partially lost in fixed-time descriptive statistics. In addition, the energy score can distinguish between predictions with identical marginal distributions but different underlying generative processes, thus more accurately assessing complex features such as multi-peak epidemic dynamics. To extend its applicability, we introduce a multi-dimensional formulation that enables a comprehensive evaluation of model performance across multiple targets or locations simultaneously.

One limitation of the energy score is the increasing computational cost of the energy score as the number of trajectories increases. The second term in the energy score expression requires the calculation of the pairwise distance between all trajectories. This can become computationally challenging when the number of trajectories is large. We show in Fig. S6 that randomly sampling even a small percent of trajectories gives good agreement with the energy score found through using the full set of stochastic trajectories.

The discrimination ability of a scoring rule illustrates the differences in scores generated by forecasts of differing quality. For example, a scoring rule has low (high) discrimination ability if forecasts with a very different quality result in the same (different) scoring value (Pinson and Tastu, 2013). A proper scoring rule can still have poor discrimination ability. It has been shown that the energy score lacks discrimination ability between forecasts with different correlation structures, but discriminates well between predictions with different means or variances (Alexander et al., 2022; Pinson and Tastu, 2013; Held et al., 2017). Many of these works are aimed at the capability of the energy score to correctly identify the true underlying distribution driving the dynamics. While this is important to consider when employing the energy score for the performance analysis of epidemic forecasts, we believe that it should not limit our use of this tool. The main goal in evaluating epidemic projections and forecasts is not the specification of the true data-generating distribution, but the identification of projections that are closest to reality.

The analysis and results reported here have considered equally weighted trajectories, but the energy score formulation also allows trajectory weights to be individually defined. This flexibility opens the door to scoring strategies that adjust weights based on past performance or prioritize specific outcomes of interest. Although such modifications may shift the interpretation of the score, they could prove valuable for building ensembles or calibrating models as new information about epidemic progression becomes available.

Beyond the case study of the Flu Scenario Modeling Hub, the energy score is broadly applicable to any epidemic forecast or prediction that produces stochastic realizations. As the reporting of trajectories becomes increasingly common, the energy score should be viewed as a widely applicable tool for model evaluation, enabling the full use of information contained within individual trajectories and supporting transparent and informative assessments of epidemic predictions.

Author contributions

Conceptualization: CB, KM, GS, JTD, AV; Investigation: CB, KM, GS, MC, JTD, AV; Methodology: CB, KM, GS, JTD, AV; Software: CB; Supervision: JTD, AV; Writing—original draft: CB, AV; Analyze the results, writing—review & editing the paper: CB, KM, GS, MC, JTD, AV.

Declaration of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

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For surveillance data, we use the National Healthcare Safety Network (NHSN) Weekly Hospital Respi- 474
ratory Data (Centers for Disease Control and Prevention, 2025) for confirmed influenza hospital admissions. 475
We use a formatted version of this data provided by the CDC FluSight Forecast Hub (FluSight Forecast Hub, 476
2025). Scenario model projections are made available by the Flu Scenario Modeling Hub (Scenario Model- 477
ing Hub, 2024a) and in their [GitHub repository](#). Our code for this project is publicly available on Zenodo 478
(Bay, 2025). 479

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Supplementary material

Evaluation of stochastic trajectory-based epidemic models using the energy score

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1 2023-24 Flu Scenario Modeling Hub Round

In this paper, we use scenario projections generated by modeling teams during the 2023-24 Flu Scenario Modeling Hub (SMH) projection round. For these projections, modeling teams aimed to predict the behavior of the influenza season under 6 potential future outcomes, with assumptions

	Season dominated by influenza A/H3N2. 40% VE against medically attended illness and hospitalization, drops in older age groups.	Season dominated by influenza A/H1N1. 40% VE against medically attended illness and hospitalization, similar across all age groups.
Higher than usual vaccine coverage (20% higher than in 2021-22 flu season)	Scenario A	Scenario B
Business as usual vaccine coverage (same as in 2021-22 flu season)	Scenario C	Scenario D
Lower than usual vaccine coverage (20% lower than in 2021-22 flu season)	Scenario E	Scenario F

Table S1: 2023-24 Flu Scenario Modeling Hub round. Description of the scenarios used by each modeling team in the 2023-24 flu SMH projection round. VE describes assumptions surrounding vaccine effectiveness (VE).

surrounding the vaccine coverage and dominant circulating strain during the 2023-24 influenza season. A description of these scenarios is shown in S1, and incident hospitalization projections were made for each U.S. state, the District of Columbia, and nationally from September 3, 2023 to June 1, 2024 [1]. Ten modeling teams provided projections under these assumptions, contributing 100 stochastic trajectories for each scenario and location. These projections are made prior to the onset of the peak 2023-24 influenza season, and are largely based on historical trends, without much calibration data available for current circulation patterns [1].

2 Other Scores for Model Evaluation

There are a range of known proper scores that can evaluate probabilistic predictions. Metrics applicable to probabilistic forecasts provided as quantiles and full predictive distributions have been widely used for assessing the performance of epidemic projections [2, 3].

For example, the weighted interval score (WIS) is a negatively-oriented proper score applied to $(1 - \alpha) \times 100\%$ prediction intervals. The WIS can be decomposed into three components: the width of the prediction interval such that wider intervals are penalized, and a cost if observations lie above or below the prediction interval [2]. The WIS assesses interval forecasts by analyzing the sharpness and calibration of the prediction intervals with respect to the observed data [3]. It is computed at each time point with prediction P and observed value y using the expression

$$\text{WIS}_{\alpha_{0:K}}(P, y) = \frac{1}{K + 0.5} \left(w_0 |y - m| + \sum_{k=1}^K w_k \text{IS}_{\alpha_k}(P, y) \right), \quad (1)$$

where IS_{α} corresponds to the interval score of the $(1 - \alpha) \times 100\%$ prediction interval,

$$\text{IS}_{\alpha}(P, y) = \underbrace{(u_{\alpha} - l_{\alpha})}_{\text{dispersion}} + \underbrace{\frac{2}{\alpha}(l_{\alpha} - y)\mathbf{1}(y < l_{\alpha})}_{\text{overprediction}} + \underbrace{\frac{2}{\alpha}(y - u_{\alpha})\mathbf{1}(y > u_{\alpha})}_{\text{underprediction}}, \quad (2)$$

where K is the number of prediction intervals used, m is the median of P , and l_{α} (u_{α}) is the lower bound (upper bound) of the prediction interval. The standard weights are chosen such that $w_k = \frac{\alpha_k}{2}$ and $w_0 = \frac{1}{2}$. In order to evaluate the performance of a full projection time series using the WIS, we take the average of the WIS calculated at each time point.

Alternatively, the continuous ranked probability score (CRPS) evaluates the cumulative distribution function (CDF) of probabilistic predictions [2, 4]. It is defined as

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} [F(x) - \mathbf{1}(x \geq y)]^2 dx,$$

where F is the CDF of the projections and y describes the observed values. The CRPS is a negatively-oriented proper score, and can be strictly proper if the predictive distribution has a finite first moment [5]. It is a generalization of the absolute error, which it reduces to if the projection is a point prediction [4].

Moreover, the CRPS can be approximated by the WIS using properties from the quantile score, given certain weights in the WIS expression [2, 6]. The CRPS also draws comparison to the energy score, which is a multivariate generalization of the CRPS [5]. In the univariate case, where a trajectory is of length one, the energy score and CRPS are identical. This highlights the similarities between how commonly used scoring rules evaluate epidemic projections.

3 Weighted Interval Score (WIS) Analysis

To compute the WIS of the Scenario Modeling Hub data, we first estimate the quantiles of the projections using the submitted trajectories for each model, scenario, and location. These quantiles are then used to calculate the WIS based on the corresponding prediction intervals. For consistency, we use the same prediction intervals employed in previous SMH projection rounds, which required quantile-based submissions; namely the 98%, 95%, 90%, 80%, ..., and 10% prediction intervals based on 23 submitted quantiles [2]. The WIS is calculated at individual time points, so we compute the average of these scores across all weeks to obtain the WIS for an entire time series.

To compare how the energy score and WIS evaluate probabilistic predictions with respect to one another, we examine how the two scores assess a synthetic predictive distribution at one time point. Note that the energy score calculated at a single time point is equivalent to the CRPS. We generate a negative binomial distribution as our predictive distribution with a mean of 60 and variance of 31, and generate 100 point predictions by randomly drawing from this distribution. We scan a range of observed values that span the predictive distribution, and calculate the energy score and WIS for each observed value. The WIS is found by estimating quantiles from the sampled

point predictions, where we use the prediction intervals used by previous rounds of the Scenario Modeling Hub (98%, 95%, 90%, 80%,..., 10%). In Fig. S1A, we show each score with the predictive distribution overlaid. We find that the energy score and WIS both find the lowest (optimal) score at the same observed value, corresponding to the points with the highest probability in the predictive distribution. The scores also diverge away from the optimal score at similar rates as the observed value decreases in likelihood. The energy score has slightly higher values for observations that are in the tail of the predictive distribution. However, as the number of quantiles used to calculate the WIS increases, the two scores will converge. This highlights similarities in the energy score and WIS, and shows that at individual time points, we can expect similarities in how the two scores evaluate a probabilistic prediction.

The WIS is calculated by summing terms that describe the dispersion, or width of the prediction intervals, along with penalty terms if the predictions over or underpredict the observed data. In Fig. S1 B, we show the WIS at each week of the influenza season, as well as the contribution of each term of the expression. In this example, the WIS is being largely penalized for underprediction, where the observations lie above the upper quantiles of the prediction. Similar to the energy score, the WIS is larger near the peak due to its dependence on the magnitude of the surveillance signal. We further highlight similarities between the energy score and WIS in S1C, where we show the correlation between the normalized energy score (Eq. 4 of main text) and normalized WIS. We define the normalized WIS in the same manner as the normalized energy score, where we divide the WIS (that has been averaged across the time series) by the magnitude of the surveillance data for the corresponding time period. We find a strong correlation between the normalized scores, but the correlation is not as strong as what we find with the non-normalized scores shown in Fig. 4D of the main text.

In the main text, we show how the energy score evaluates models with respect to one another. This allows us to rank models in order to see which models perform better or worse. While we find that the energy score and WIS are highly correlated, we are also interested in whether the two scores rank models similarly as well. In Fig. S1D, we show a histogram of the Kendall's τ rank correlation for the ranking of models for each scenario and location in the 2023-24 Flu SMH hospitalization predictions. A rank correlation of 1 means that the energy score and WIS rank models identically. We find that the rankings are very similar, with a mean rank correlation across models and scenarios of 0.87. This further highlights the similarities in how the WIS and energy score evaluate model performance, but shows that there are cases where the model rankings differ.

Fig. S2 shows the correlation between the normalized energy score and normalized WIS separated for each scenario for the 'MOBS_NEU-GLEAM_FLU' model 2023-24 Flu SMH projections. We find that the correlations between scores are stronger for Scenarios A, C, and E, which correspond to the scenarios assuming an Influenza A/H3N2 dominant season. Examining the trajectories of this model for each scenario (Fig. 3 of main text), we find that the A/H3N2 dominant scenarios (A,C,E) exhibit simple behavior, with a single peak occurring around the same time, with uncertainty in the peak magnitude. The A/H1N1 dominant scenarios (B,D,F) have more complex dynamics, with one or more peaks occurring at different times. This relates to the discussion in the main text related to how the energy score and WIS behave in situations with multi-peak dynamics (Fig. 2 of main text). We see that in the scenarios with simpler dynamics, the correlation between scores is higher than scenarios with variability in the peak timing. This is likely due to the same phenomenon we discuss in the main text, where the energy score and WIS assess complex multi-peak dynamics differently.

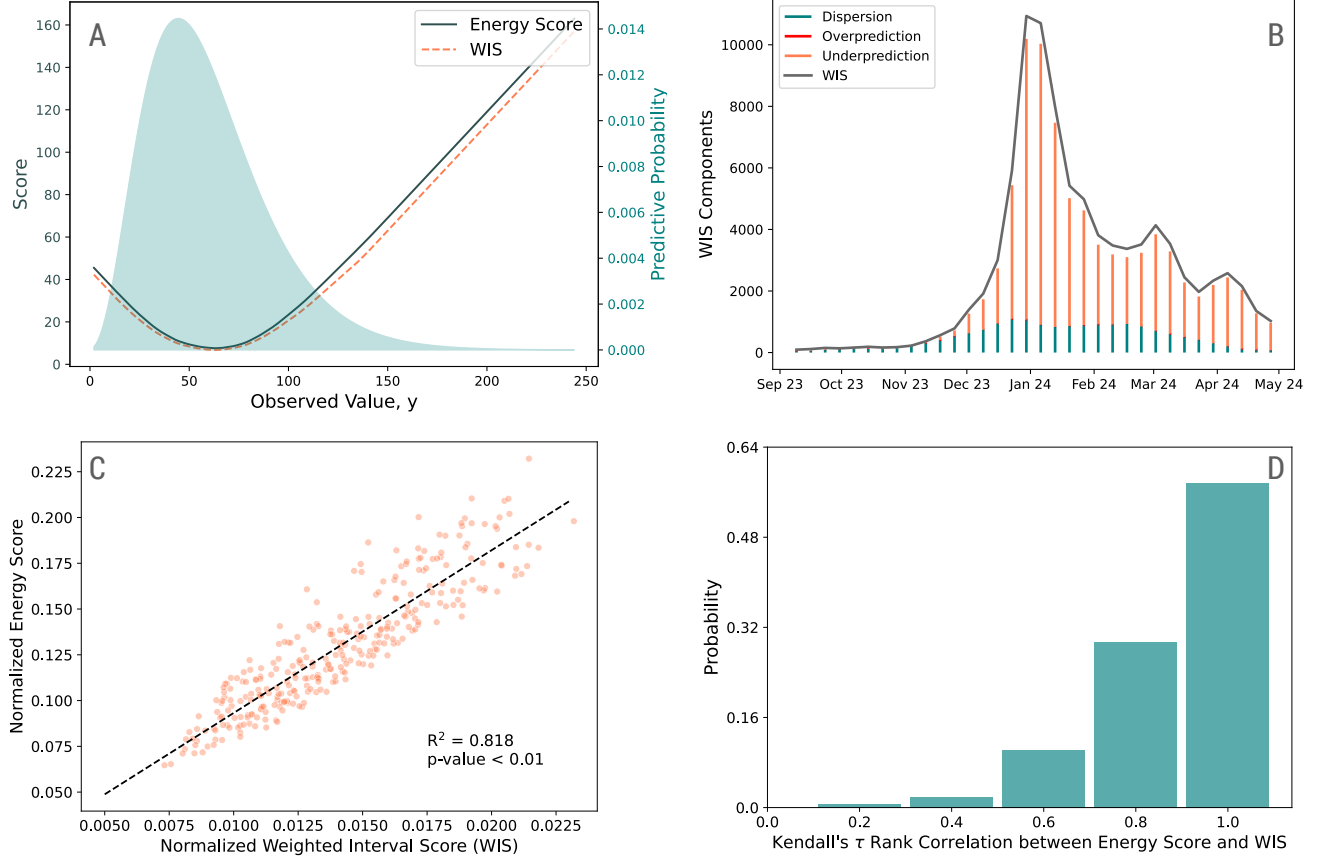


Figure S1: (A) Decomposition of the WIS terms at each week for the 'MOBS_NEU-GLEAM.FLU' model 2023-24 Flu SMH incident hospitalization projections for Scenario D, where blue show the contribution of the dispersion term, red describes the penalty for overprediction, and orange shows the penalty for underprediction. The gray line shows the full score calculated at each week. (B) Comparison of the energy score (gray) and WIS (dashed orange) calculated at one time point as a function of an observed value y given an underlying predictive distribution (blue) with 100 samples drawn from the predictive distribution to calculate the scores. The predictive distribution is a negative binomial distribution with a mean of 60 and variance of 31. (C) Relationship between the normalized WIS and normalized energy score for the 'MOBS_NEU-GLEAM.FLU' model 2023-24 Flu SMH projections. Each point describes the normalized scores for a location and scenario. The WIS was found by estimating the quantiles estimated from the trajectories. Both scores are normalized by dividing the score by the sum of the surveillance data. Dashed line shows the linear regression describing the relationship between the scores. (D) Histogram of the Kendall's τ rank correlation between the model rankings for the energy score and WIS for each scenario and location for the 2023-24 Flu SMH incident hospitalization projections. A Kendall's τ rank correlation of 1 means that the energy score and WIS rank models identically.

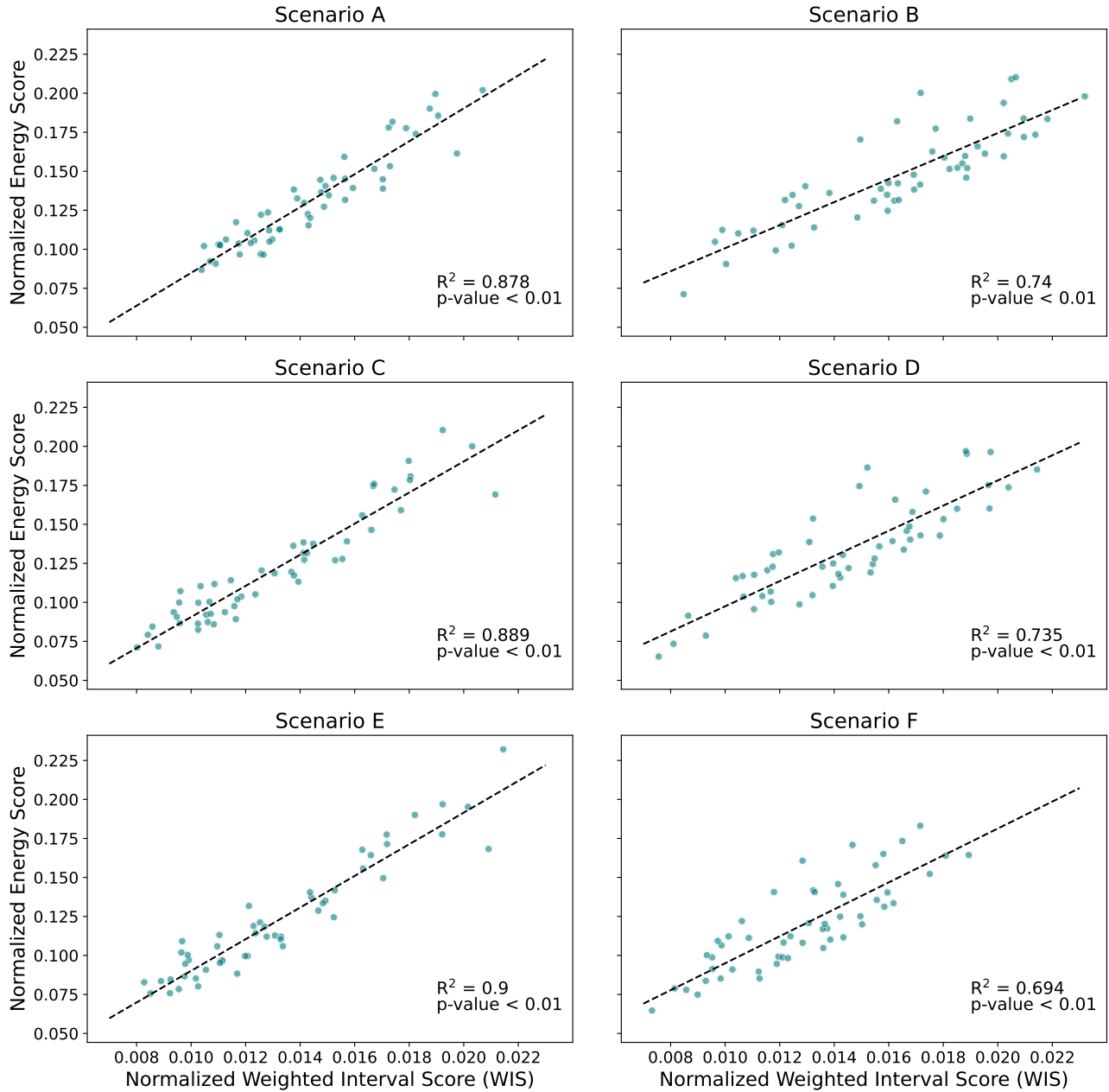


Figure S2: Correlation between the normalized energy score and normalized WIS for each scenario for the ‘MOBS-NEU-GLEAM-FLU’ model 2023-24 Flu SMH incident hospitalization projections. Each point represents the corresponding scores for a single location for a given scenario. Both scores are normalized by dividing the score by the sum of the surveillance data.

4 Naive baseline model

While scoring metrics provide a value against which we can compare models within the Scenario Modeling Hub, this does not actually tell us how well the scenario projections match reality. For example, we can say that a projection with an energy score of 10 performs better than a projection

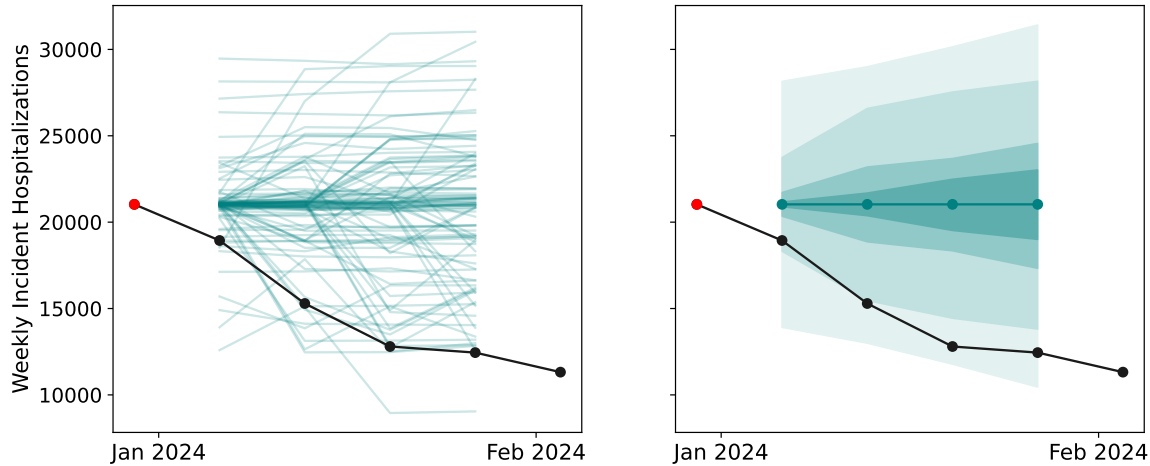


Figure S3: Example of the baseline forecast model for 1-4 week ahead projections given a most recently observed data point of December 30, 2024 (red). Trajectories are generated from the baseline model (left) and can be transformed into quantiles (right). Black line shows observed surveillance data.

with an energy score of 20, but we do not know whether the energy score of 10 is actually a ‘good’ prediction. Therefore, we employ a naive baseline forecasting model as a reference point against which we compare the performance of the Flu SMH scenario projections. We use the baseline model framework described by Ray et al. where the median is equal to the last observed data point, and the uncertainty around the median is described by historical differences in the changes in incidence counts between consecutive weeks [7]. This creates flat projections of the most recent data, with increasing uncertainty for more distant horizons. Code for the naive baseline model is adapted from the CovidEnsembles (<https://github.com/reichlab/covidEnsembles/>) and CovidModels (<https://github.com/reichlab/covidModels/>) packages in R. To generate trajectories for this model, we create the first horizon (ex. 1-week-ahead) by summing samples drawn from the historical distribution of weekly differences with the most recently observed data point. For longer horizons, the weekly differences are sampled again and summed to the previous horizon projected incidence counts, which generates the expanding uncertainty we observe [7]. We repeat this for the number of weeks ahead being predicted, where the difference from one week to the next is randomly sampled from the distribution of historical differences in incidence counts. Fig. S3 shows an example of the baseline model forecast trajectories and corresponding quantile format.

With this framework, we generate trajectories for 4-week-ahead baseline forecasts for the 2023-24 influenza season in the U.S. We use historical data starting in September 2022 to estimate historical changes in incidence across weeks. Using these trajectories, we calculate the energy score for each U.S. state for the 2023-24 flu season with the 4-week-ahead naive baseline model. In order to have baseline forecasts for each date predicted by the SMH projections, the 4-week-ahead forecasts begin August 12, 2023. In the main text, we compare these scores with the scenario projections to assess whether long-term scenario projections perform better or worse than a short-term baseline forecast.

	No. locations predicted	Minimum	5% quantile	Median	95% quantile	Maximum
Scenario A	52	0.87	0.91	1.16	1.49	1.59
Scenario B	52	0.73	0.84	1.29	1.87	2.07
Scenario C	52	0.76	0.79	1.06	1.41	2.16
Scenario D	52	0.66	0.78	1.20	1.76	1.89
Scenario E	52	0.80	0.81	1.06	1.45	2.42
Scenario F	52	0.68	0.74	1.05	1.55	1.91

Table S2: Descriptive statistics of the energy score ratio for the ‘MOBS_NEU-GLEAM_FLU’ model projections. Table showing the number of locations predicted, minimum, 5% quantile, median, 95% quantile and maximum for energy score ratios across locations for influenza hospitalization predictions for the ‘MOBS_NEU-GLEAM_FLU’ model of the 2023-24 SMH round across locations for each scenario. The energy score ratio is calculated by dividing the energy score of the SMH model at each scenario and location by the energy score of a 4-week-ahead naive baseline model at each location.

5 Energy Score Ratio Descriptive Statistics

In Table S2 we show descriptive statistics of the energy score ratio for the ‘MOBS_NEU-GLEAM_FLU’ model hospitalization projections across all locations for each scenario of the 2023-24 Flu SMH round. This corresponds to the energy score ratio data used in Fig. 4E in the main text. The energy score ratio compares the energy score of the ‘MOBS_NEU-GLEAM_FLU’ model with a 4-week-ahead naive baseline model at each scenario and location. An energy score ratio less than 1 corresponds to cases where the ‘MOBS_NEU-GLEAM_FLU’ model performed better than the baseline model.

In Table S3 we show descriptive statistics of the energy score ratio for each scenario and model across all locations for incident hospitalization projections of the 2023-24 Flu SMH round. This corresponds to the data used in Fig. 5 in the main text, which describes the energy score ratio comparing the energy score of each SMH model to the energy score of a 4-week-ahead naive baseline model. We only show data for the models we used in our analysis, excluding models submitted to the SMH that only reported projections for a single location. Note that there are 52 possible locations included in our analysis: the 50 U.S. states, District of Columbia, and US national projections.

In Table S4, we show descriptive statistics of the energy score ratio for the ensemble model generated by bundling the trajectories of each model together with equal weight. This describes the data used in Fig. 6B in the main text. The energy score ratio compares the behavior of this ensemble with the 4-week-ahead naive baseline model. We include trajectories for all 10 individual models that submitted projections to the 2023-24 Flu SMH, including those that only report scenario projections for a single location.

Label	Model	No. locations predicted	Minimum	5% quan- tile	Median	95% quantile	Maximum
Scenario A	Model I	52	0.87	0.91	1.16	1.49	1.59
	Model II	13	0.50	0.50	0.84	1.13	1.13
	Model III	52	0.48	0.60	1.55	3.40	7.04
	Model IV	52	0.51	0.65	1.80	4.90	7.48
	Model V	52	0.93	1.09	1.79	2.52	4.70
	Model VI	51	0.66	0.87	3.19	12.22	16.66
Scenario B	Model I	52	0.73	0.84	1.29	1.87	2.07
	Model II	13	0.79	0.81	1.02	1.55	1.61
	Model III	52	0.51	0.56	0.94	2.59	4.49
	Model IV	52	0.39	0.51	1.71	3.78	4.73
	Model V	52	0.88	1.01	1.70	2.42	4.50
	Model VI	51	0.72	0.89	3.03	11.72	16.70
Scenario C	Model I	52	0.76	0.79	1.06	1.41	2.16
	Model II	13	0.51	0.57	1.44	4.20	5.86
	Model III	52	0.54	0.66	1.79	3.91	7.87
	Model IV	52	0.50	0.79	2.31	6.02	9.35
	Model V	52	0.93	1.09	1.80	2.52	4.71
	Model VI	51	0.68	0.89	2.93	12.19	16.52
Scenario D	Model I	52	0.66	0.78	1.20	1.76	1.89
	Model II	13	0.48	0.51	0.76	2.74	3.40
	Model III	52	0.44	0.52	1.03	2.93	5.16
	Model IV	52	0.28	0.58	2.01	4.51	5.77
	Model V	52	0.88	1.01	1.70	2.42	4.50
	Model VI	51	0.65	0.82	3.07	11.84	16.58
Scenario E	Model I	52	0.80	0.81	1.06	1.45	2.42
	Model II	13	0.45	0.71	2.75	7.51	10.00
	Model III	52	0.50	0.72	2.03	4.48	8.75
	Model IV	52	0.59	1.17	2.95	7.37	11.42
	Model V	52	0.93	1.09	1.79	2.52	4.69
	Model VI	51	0.69	0.81	3.09	11.75	16.27
Scenario F	Model I	52	0.68	0.74	1.05	1.55	1.91
	Model II	13	0.45	0.81	1.94	5.35	6.23
	Model III	52	0.51	0.54	1.22	3.36	6.09
	Model IV	52	0.26	0.86	2.35	5.46	6.74
	Model V	52	0.88	1.01	1.70	2.42	4.50
	Model VI	51	0.71	0.84	3.07	11.95	16.91

Table S3: Descriptive statistics of the energy score ratio for the all analyzed models and scenarios. Table showing the number of locations predicted, minimum, 5% quantile, median, 95% quantile and maximum for energy score ratios across locations for influenza hospitalization predictions for each scenario and model of the 2023-24 SMH round. The energy score ratio is calculated by dividing the energy score of the SMH model at each scenario and location by the energy score of a 4-week-ahead naive baseline model at each location.

	No. locations predicted	Minimum	5% quantile	Median	95% quantile	Maximum
Scenario A	52	0.57	0.63	1.09	2.23	2.34
Scenario B	52	0.55	0.66	1.04	1.72	1.97
Scenario C	52	0.59	0.65	1.17	2.42	2.85
Scenario D	52	0.55	0.59	1.02	1.93	2.01
Scenario E	52	0.61	0.65	1.26	2.74	3.42
Scenario F	52	0.52	0.58	1.08	2.13	2.33

Table S4: Descriptive statistics of the energy score ratio for the ensemble model projections. Table showing the number of locations predicted, minimum, 5% quantile, median, 95% quantile and maximum for energy score ratios across locations for influenza hospitalization predictions for the ensemble of trajectories model across locations for each scenario. The energy score ratio is calculated by dividing the energy score of the ensemble model at each scenario and location by the energy score of a 4-week-ahead naive baseline model at each location.

6 Multi-Dimensional Energy Score

To calculate the multi-dimensional energy score, we must group trajectories across the different dimensions to create trajectory matrices, where each matrix has one time-series trajectory for each target outcome dimension. The choice of this grouping changes the resulting energy score value. Therefore, trajectories within each matrix should be paired such that trajectories generated by the same simulations are grouped together. While the SMH data is reported with trajectory identifiers, it cannot be ensured that the trajectories across locations are paired. We perform a sensitivity analysis when calculating the multi-dimensional energy score on this data to study the effect of grouping unpaired trajectories. We run multiple iterations of randomizing how these trajectories are paired into the trajectory matrices. In Table S5, we show statistics describing 50 iterations of randomizing the matrices for calculating the multi-dimensional energy score for ‘MOBS_NEU-GLEAM_FLU’ model for all locations, shown for each scenario. We see that the standard deviation and range of the energy score values is small, with a standard deviation of 0.0002-0.0004 and a range of 0.001-0.002. This suggests that randomly pairing trajectories does not greatly impact the energy score. Moreover, this means that using paired trajectories for the multi-dimensional energy score may not be required if this information is not available. In this example, we also see that it does not impact the ranking of the scenarios, as none of the scores for each scenario overlap with one another.

In the multi-dimensional energy score expression, we introduce a normalization factor, Φ^2 , which ensures that outcome dimensions with larger magnitude do not dominate the multi-dimensional energy score value. However, it can be useful to keep the emphasis on outcome dimensions (i.e. locations) with larger signals such that poorly predicting a larger signal is penalized more than dimensions with smaller signals. We compare results of the multi-dimensional energy score with and without the normalization factor, Φ_j^2 , in Fig. S4. To calculate the multi-dimensional energy score with no normalization factor, we simply remove Φ_j^2 from the expression (i.e. $\Phi_j^2=1$) in Eq. 5 of the main text. Similar to Fig. 4E of the main text, we compute the non-normalized multi-

	Mean	Standard Deviation	Minimum	Maximum	Range
Scenario A	0.868	0.0003	0.867	0.868	0.002
Scenario B	1.005	0.0002	1.004	1.005	0.001
Scenario C	0.843	0.0004	0.841	0.843	0.002
Scenario D	0.917	0.0002	0.917	0.918	0.001
Scenario E	0.857	0.0004	0.856	0.858	0.002
Scenario F	0.816	0.0003	0.815	0.817	0.001

Table S5: Descriptive statistics of the multi-dimensional energy score for the ‘MOBS_NEU-GLEAM_FLU’ model. Table showing the mean, standard deviation, minimum, maximum, and range for the multi-dimensional energy score across locations for the ‘MOBS_NEU-GLEAM_FLU’ model influenza hospitalization predictions for each scenario of the 2023-24 SMH round, given 50 iterations of randomly pairing trajectories.

dimensional energy score for 50 iterations of randomly shuffling the trajectory identifiers used to pair trajectories, and create an energy score ratio comparing the ‘MOBS_NEU-GLEAM_FLU’ model to the 4-week-ahead naive baseline model. The blue star in Fig. S4 represents the median non-normalized multi-dimensional energy score ratio, which can be compared to the red star, which shows the multi-dimensional score with normalization included. We find that the energy score ratios are quite similar, and the deviations from the normalized score are likely a consequence of scores from large states biasing the calculation.

7 Scenario Modeling Hub Ensemble Model Comparison

In the main text, we describe a method of generating an ensemble model where we simply bundle all trajectories of component models and assign equal weight to each one. The SMH reports 3 ensemble models with projections of the submitted modeling teams. These ensemble models are generated using the Vincent average and linear opinion pool (LOP) methods, where the LOP method is constructed in both a trimmed, where extreme values are excluded, and untrimmed manner [3]. The 3 resulting ensemble models are called the Ensemble_vincent, Ensemble_LOP, and Ensemble_LOP_untrimmed models. These models are calculated and reported in quantile format. We are interested in comparing the difference between the SMH ensemble models and the ensemble of trajectories model analyzed in the main text. Since the SMH ensemble models are only reported as quantiles (and not trajectories), we use the WIS for this analysis. In Fig. S5, we show the relationship between the WIS of each SMH ensemble model and the WIS of the ensemble of trajectories model. The WIS for the trajectory ensemble is calculated by estimating quantiles from the trajectories before computing the WIS. We find a strong positive correlation between the trajectories ensemble and all three SMH ensemble models. However, the Ensemble_LOP_untrimmed model has the strongest correlation with the ensemble of trajectories method. This is reasonable since both methods account for the full uncertainty of all projections, where the Ensemble_vincent and Ensemble_LOP models reduce this variation. This demonstrates the trajectories ensemble performance in

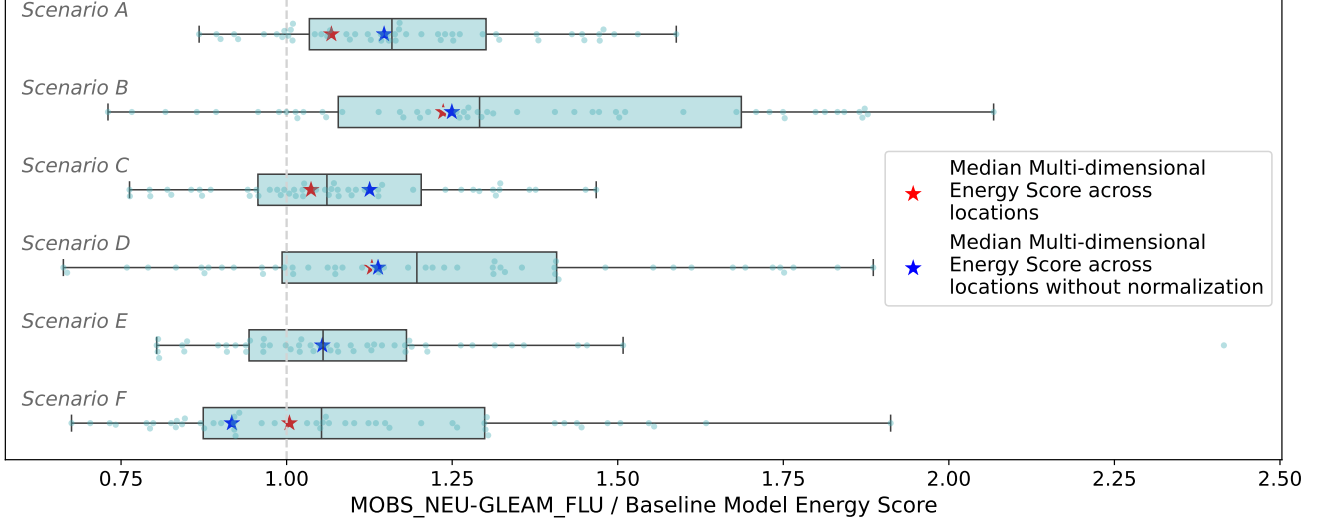


Figure S4: Boxplot of energy score ratio across 52 locations for the ‘MOBS_NEU-GLEAM_FLU’ model compared to a 4-week-ahead naive baseline model for each scenario. Vertical dashed line shows where the ‘MOBS_NEU-GLEAM_FLU’ and baseline model have the same scores, where ratios below one describe when the model performs better than the baseline. The overlaid red stars show the median of the multi-dimensional energy score ratio across locations for 50 iterations of randomizing the trajectory pairings. The blue stars show the multi-dimensional energy score without the use of a normalization factor. Boxplots are created such that the box shows the 25%, 50% and 75% quantiles, and the whiskers represent $1.5 \times \text{IQR}$.

comparison to the SMH ensemble models. Choosing different weighting of trajectories within the ensemble of trajectories could change this uncertainty and allow for many other variations of using trajectories for ensemble model generation.

8 Sampling trajectories in energy score calculation

Given the pairwise comparison between all pairs of trajectories in order to calculate the second term of the energy score, this computation could become infeasible if we are looking at thousands or millions of trajectories from a model’s output. We propose the use of sampling from the full group of trajectories if it is intractable to find the energy score using all of them. We show the outcome of sampling trajectories using the ‘MOBS_NEU-GLEAM_FLU’ model for the 2023-24 Round 1 of the Flu Scenario Modeling Hub (SMH). To do this, we randomly sample n trajectories for each location and scenario out of the 100 trajectories submitted, and we calculate the energy score using these n trajectories. We repeat this process for 50 iterations and compare the results with the energy score found by using the full set of trajectories.

In Fig. S6 A, we show how the number of sampled trajectories used in the energy score calculation impacts how close our estimated energy score is to the true value. This is shown across 50 iterations of sampling trajectories for four different densities of samples. We find good agreement even for very low number of trajectories sampled, and just 10% of trajectories is sufficient to get a near perfect estimation of the true energy score, with an $R^2 = 0.981$. If we average across these

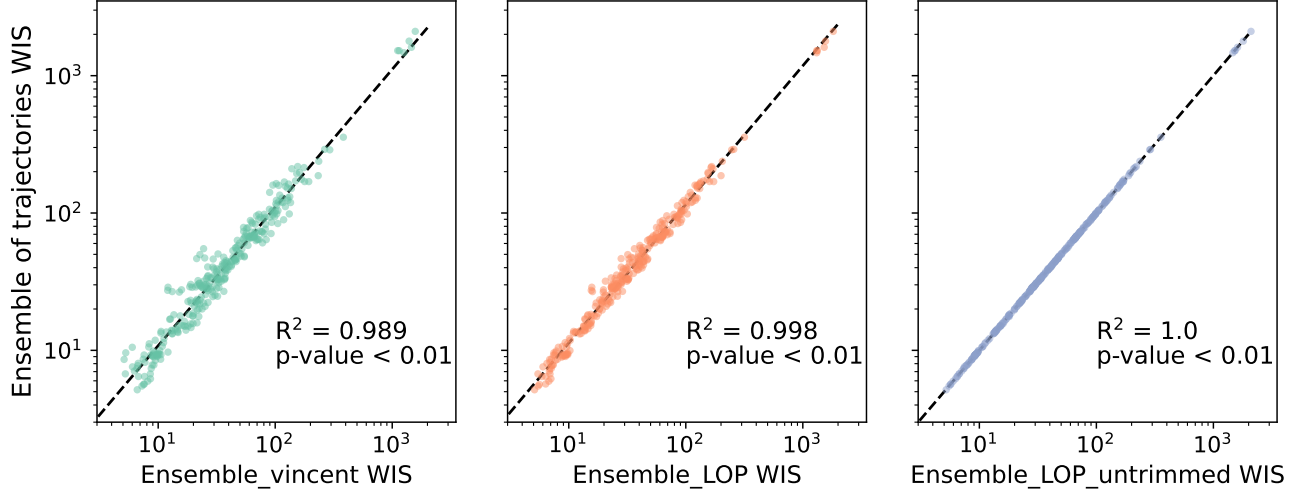


Figure S5: Correlation between the WIS of the ensemble models reported by the SMH (x-axis) and the WIS of the ensemble of trajectories method (y-axis). Each subplot shows one of the 3 different SMH-reported ensemble models with Ensemble_vincent (left), Ensemble_LOP (middle), and Ensemble_LOP_trimmed (right). Each point represents the corresponding WIS scores for a given scenario and location. Dashed black line shows the linear regression fit, with the R^2 and two-sided p-value describing the correlation.

iterations, we can improve our estimation for a lower percentage of trajectories sampled. Fig. S6 B shows that with only 5% of trajectories used in the energy score calculation, we get an energy score nearly identical to the truth. This highlights the utility of sampling in calculating the energy score when the total number of stochastic trajectories may be computationally challenging. Even when taking a small sample of trajectories, we can find a close estimate of the true energy score.

9 Proofs of properness

In this section we will show proofs that the energy score is a strictly proper scoring rule and that the weighted interval score (WIS) is a proper score. Let us begin with a few important definitions that will be used in both proofs. Let $S(F, y)$ be a score for a probability distribution function F that gives probabilistic predictions of some target x . Let y be the observed surveillance data of the prediction target of interest, and G the true underlying distribution for y . We call $E_G[S(F, y)] = S(F)$ the expected score. A negatively-oriented score is such that smaller scores correspond to better predictions. Our definitions of the energy score and WIS are both negatively-oriented scores. Given these definitions, we can say that S is a negatively-oriented proper score if $S(G) \leq S(F)$ for all F . Moreover, S is a negatively oriented strictly proper score if $S(G) \leq S(F)$ for all F , with equality if and only if $F = G$. Therefore, to prove that a score S is proper, we must show that $S(F) - S(G) \geq 0$. To prove that a score is strictly proper, we must show that $S(F) - S(G) \geq 0$ with equality if and only if $F = G$.

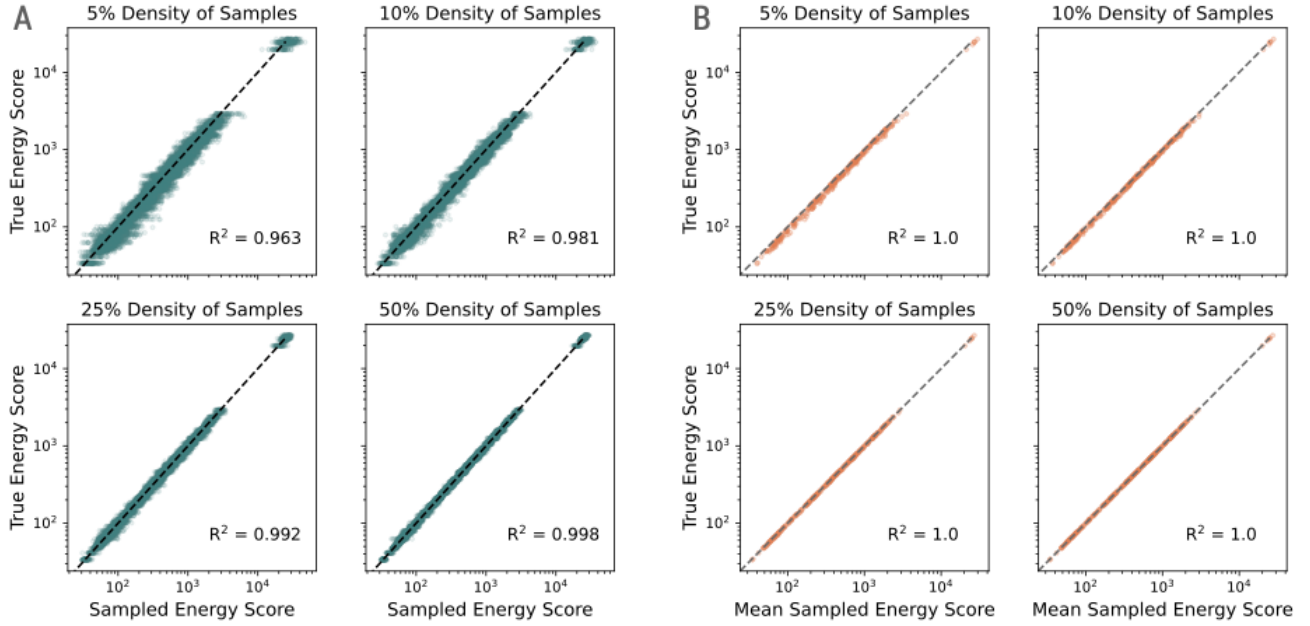


Figure S6: (A) Sampled energy score compared to the true energy score for different densities of sampled trajectories (5, 10, 25, and 50 %) for the ‘MOBS_NEU-GLEAM_FLU’ model. Dots show a single sampled energy score compared to its corresponding true value for a given location and scenario. (B) Sampled energy score averaged across 50 iterations compared to the true energy score for different densities of sampled trajectories (5, 10, 25, and 50 %) for the ‘MOBS_NEU-GLEAM_FLU’ model. Dots show the average sampled energy score compared to its corresponding true value for a given location and scenario. Black dotted line shows the $y=x$ line where the sampled and true energy scores are equivalent.

Proof that the energy score is strictly proper

To show that the energy score $ES(F, y)$ is strictly proper, we need to show that the expected energy score is uniquely minimized when the prediction distribution F equals the distribution G of the observed value. More formally, we want to show that

$$E_G[ES(F, y)] \equiv ES(F) \geq ES(G) \quad \forall F \quad (3)$$

where the equality holds if and only if $F = G$.

From Refs. [5, 8] we can show that for y a d -dimensional vector, the energy score is

$$ES(F, y) = A \int_{\mathbb{R}^d} \frac{\|\phi_F(v) - e^{iv \cdot y}\|^2}{\|v\|^{d+1}} dv, \quad (4)$$

where $\phi_F(v) \equiv E_F[e^{iv \cdot x}]$ is the characteristic function of F , v is also a d -dimensional vector, $\|\cdot\|$ is the Euclidean distance, and

$$A = -\frac{\Gamma\left(\frac{d+1}{2}\right)}{2\pi^{\frac{d+1}{2}}}.$$

This shows that the energy score calculates the distance between the characteristic function between the prediction F and the characteristic function of the observed data [5]. Given this, the expected energy score corresponds to:

$$E_G [\text{ES}(F, y)] = A \cdot E \left[\int_{\mathbb{R}^d} \frac{||\phi_F(v) - e^{iv \cdot y}||^2}{||v||^{d+1}} dv \right] . \quad (5)$$

To simplify the numerator, we use the fact that the squared norm of a complex number z is given by $||z||^2 = z\bar{z}$, where $\bar{z}$ is the complex conjugate of z . If we apply this to our expression of the expected energy score we find

$$= A \cdot E \left[\int_{\mathbb{R}^d} \frac{(\phi_F(v) - e^{iv \cdot y})(\bar{\phi}_F(v) - e^{-iv \cdot y})}{||v||^{d+1}} dv \right] .$$

If we multiply through all terms and take the expectation over y we get:

$$\begin{aligned} &= A \cdot E \left[\int_{\mathbb{R}^d} \left(\frac{\phi_F(v)\bar{\phi}_F(v) - e^{iv \cdot y}\bar{\phi}_F(v) - e^{-iv \cdot y}\phi_F(v) + 1}{||v||^{d+1}} \right) dv \right] , \\ &= A \int_{\mathbb{R}^d} \left(\frac{\phi_F(v)\bar{\phi}_F(v) - \phi_G(v)\bar{\phi}_F(v) - \bar{\phi}_G(v)\phi_F(v) + 1}{||v||^{d+1}} \right) dv , \end{aligned}$$

where $\phi_G(v)$ is the characteristic function of G describing the observed data. If we add and subtract $\phi_G(v)\bar{\phi}_G(v)$ from the numerator inside the integral, we can separate the integral in two parts, to find

$$\begin{aligned} &= A \int_{\mathbb{R}^d} \left(\frac{\phi_F(v)\bar{\phi}_F(v) - \phi_G(v)\bar{\phi}_F(v) - \bar{\phi}_G(v)\phi_F(v) + \phi_G(v)\bar{\phi}_G(v)}{||v||^{d+1}} \right) dv \\ &\quad + A \int_{\mathbb{R}^d} \frac{1 - \phi_G(v)\bar{\phi}_G(v)}{||v||^{d+1}} dv , \end{aligned}$$

If we gather the terms and use the complex conjugate definition of the norm, we can rewrite the numerator as a squared Euclidean distance and show that the expected energy score is

$$E_G [\text{ES}(F, y)] = A \int_{\mathbb{R}^d} \frac{||\phi_F(v) - \phi_G(v)||^2}{||v||^{d+1}} dv + B , \quad (6)$$

where $B = A \int_{\mathbb{R}^d} \frac{1 - \phi_G(v)\bar{\phi}_G(v)}{||v||^{d+1}} dv$ is the second integral described above. Since B does not depend on the predictions F , we must only look at the first term in order to minimize the expected energy score. This shows that the expected energy score is minimized if and only if $\phi_F(v) = \phi_G(v)$. Since the characteristic functions $\phi_F(v)$ and $\phi_G(v)$ uniquely characterize the underlying probability distributions, this condition is equivalent to $F = G$, which proves the energy score is strictly proper. $\square$

Proof that WIS is proper

For a prediction composed of quantiles, we can say that the expected score given the predicted quantiles r_i and probability distribution of the observed data G is

$$E_G[S(r_1, r_2; y)] = S(r_1, \dots, r_k; G).$$

We draw comparison to a general quantile scoring rule to show that the WIS is proper [5].

Ref. [5] shows that for predicting a single quantile, the scoring rule

$$S(r; y) = \alpha s(r) + (s(y) - s(r))\mathbb{1}(y \leq r) + h(y) \quad (7)$$

is proper for quantile levels $\alpha \in (0, 1)$, s is a non-decreasing functions describing the score, r is the predicted quantile, and $h(y)$ is an arbitrary function. First, we will show that a score of this form is proper, using Theorem 6 from Ref. [5] and then we show how the interval score can be written in a general form of this example.

Now we will show that Eq. 7 is proper for s non-decreasing. To do this, we need to show that for a score $S(r; y)$ with predicted quantile r , the expected score is maximized by the true quantile q of G at level α . In this proof, we show the case for a score S that is positively-oriented, where larger scores are better. This means that the score is proper if it is maximized by the underlying distribution of the observed values. The same proof can be shown for a negatively-oriented score by multiplying all terms by negative one. The expected score can be found by calculating

$$S(r_1, r_2; G) = \int S(r_1, r_2; y) \partial \mu_G(y),$$

where $\mu_G(y)$ this probability measure for the observed data distribution G [5]. Let the probability measure be such that $\mu_G(q) = \alpha$. Therefore, a scoring rule in the form of Eq. 7 is proper if $S(q; G) \geq S(r; G)$. In this example, we assume that $r < q$. The score $S(r; y)$ can be written as

$$S(r; y) = \alpha s(r) + (s(y) - s(r))\mathbb{1}(y \leq r) + h(y). \quad (8)$$

Therefore, the expected score given predicted quantiles r is

$$S(r; G) = \int_0^\infty [\alpha s(r) + (s(y) - s(r))\mathbb{1}(y \leq r) + h(y)] \partial \mu_G(y),$$

if we consider $y \in [0, \infty)$. We take the integral of the term multiplied by the indicator function from 0 to r due to the indicator function restriction, simplify, and find

$$S(r; G) = \alpha s(r) - s(r)\mu_G(r) + \int_0^r s(y) \partial \mu_G(y) + \int_0^\infty h(y) \partial \mu_G(y).$$

To show that the true quantile maximizes this score, we subtract this expression from the score calculated using the true quantiles q . Using the same methodology as above, the expected score of the true quantiles is

$$S(q; G) = \alpha s(q) - s(q)\mu_G(q) + \int_0^q s(y) \partial \mu_G(y) + \int_0^\infty h(y) \partial \mu_G(y) = \int_0^q s(y) \partial \mu_G(y) + \int_0^\infty h(y) \partial \mu_G(y).$$

The first two terms cancel since $\mu_G(q) = \alpha$. Now, if we subtract the expected score of the true and predicted quantiles, we find

$$S(q; G) - S(r; G) = \int_0^q s(y) \partial \mu_G(y) + \int_0^\infty h(y) \partial \mu_G(y) - \left[\alpha s(r) - s(r) \mu_G(r) + \int_0^r s(y) \partial \mu_G(y) + \int_0^\infty h(y) \partial \mu_G(y) \right],$$

by canceling the $h(y)$ term and rearranging terms, we get

$$S(q; G) - S(r; G) = s(r) \mu_G(r) - \alpha s(r) + \int_0^q s(y) \partial \mu_G(y) - \int_0^r s(y) \partial \mu_G(y).$$

Now we can combine the integrals since $r < q$, and find that

$$S(q; G) - S(r; G) = s(r) \mu_G(r) - \alpha s(r) + \int_r^q s(y) \partial \mu_G(y).$$

Since $s(y)$ is non-decreasing, then $s(y) \geq s(r)$ for $y \in [r, q]$. Therefore, $\int s(y) d\mu_G(y) \geq \int s(r) \partial \mu_G(y) = s(r)(\mu_G(q) - \mu_G(r))$. Using this and the fact that $\mu_G(q) = \alpha$,

$$S(q; G) - S(r; G) \geq s(r)(\mu_G(q) - \mu_G(r)) + s(r) \mu_G(r) - \alpha s(r) = 0.$$

Therefore, $S(q; G) \geq S(r; G)$, meaning that the expected score is maximized by the observed values. Note that this is the case for $r < q$, but a similar proof would follow for the alternative case. This shows that a score of the form Eq. 8 is a proper score, since the expected score is maximized by the true quantiles. To translate this to a case with a negatively-oriented score, we simply must negate the expression above and show that the score is minimized by its true quantiles.

This score can be written in a more general case for predicting k quantiles at levels $\alpha_1, \dots, \alpha_k \in (0, 1)$ using the expression

$$S(r_1, \dots, r_k; y) = \sum_{i=1}^k [\alpha_i s_i(r_i) + (s_i(y) - s_i(r_i)) \mathbb{1}(y \leq r_i)] + h(y), \quad (9)$$

where it is proper if s_i are non-decreasing [5]. This proof is an extension of the proof shown above for predicting multiple quantiles.

We can write the interval score in the form presented in Ref. 9 where $\alpha_1 = \frac{\alpha}{2}$, $\alpha_2 = 1 - \frac{\alpha}{2}$, $s_1(y) = s_2(y) = 2\frac{y}{\alpha}$, and $h(y) = -2\frac{y}{\alpha}$, as described in [5]. Note that this will give us the negated version of the typical, negatively oriented interval score [2]. Plugging in these functions into Eq. 9, we find that the interval score is

$$IS = r_1 + \frac{2}{\alpha}(y - r_1) \mathbb{1}(y \leq r_1) + 2\frac{r_2}{\alpha} - r_2 + \frac{2}{\alpha}(y - r_2) \mathbb{1}(y \leq r_2) - 2\frac{y}{\alpha}.$$

Rearranging terms and simplifying, we get

$$IS = (r_1 - r_2) + \frac{2}{\alpha}(y - r_1) \mathbb{1}(y \leq r_1) + \frac{2}{\alpha}(r_2 - y) \mathbb{1}(y > r_2).$$

To get the expression for the interval score, we let $r_1 = l$, $r_2 = u$ the lower and upper quantiles of the prediction, respectively. This can be written as

$$IS = (l - u) + \frac{2}{\alpha}(y - l)\mathbb{1}(y \leq l) + \frac{2}{\alpha}(u - y)\mathbb{1}(y > u).$$

This shows that the interval score can be written in the generalized form of Eq. 9, meaning that it is a proper score. This is the positively-oriented formulation of the interval score, but to write the typical, negatively-oriented expression we must multiply all terms by -1. Since the weighted interval score is a linear combination of interval scores, it is also proper because a sum across proper scores is still proper [2].

References

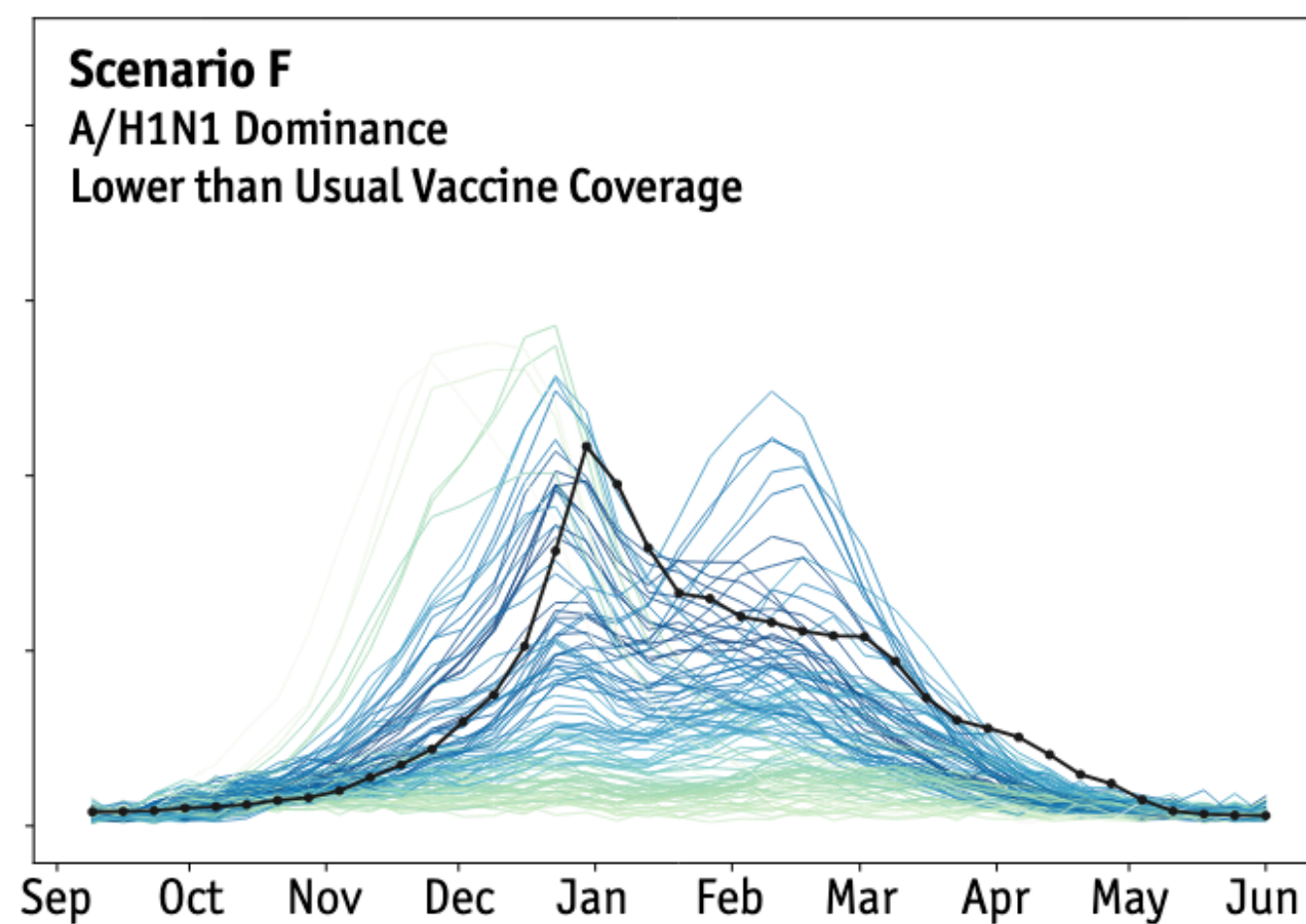
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Evaluating Stochastic Trajectory-Based Epidemic Models Using the Energy Score

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Trajectory-Based Evaluation

A performance metric is needed that fully captures the information contained in models' stochastic trajectories, beyond what quantile summaries can provide.



Energy Score Calculation: Observations vs. Trajectories

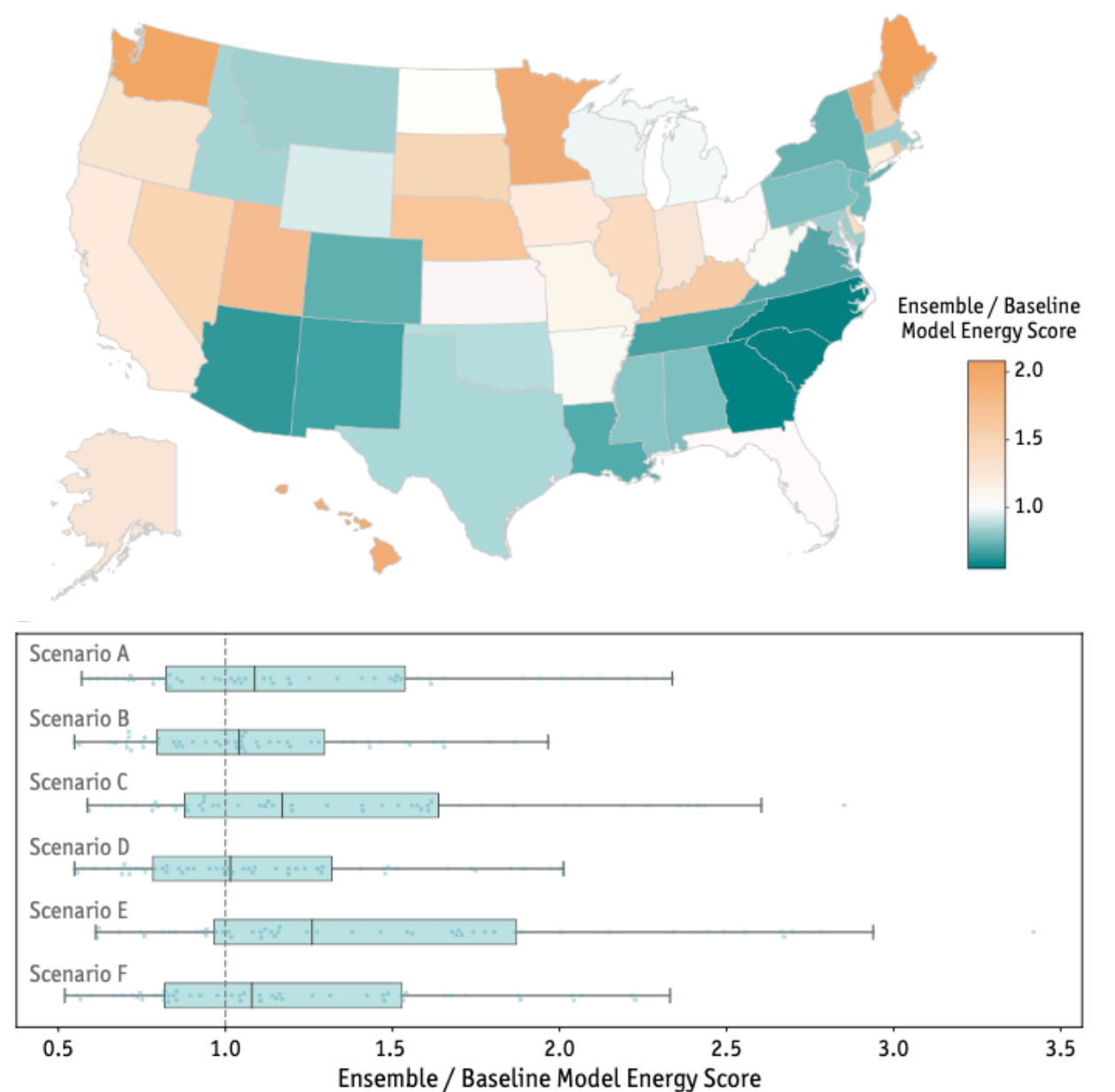
The energy score provides a principled framework for evaluating epidemic model outputs in trajectory format, avoiding the need to summarize stochastic realizations into quantiles.

$$ES = \text{distance}(\text{traj} \leftrightarrow \text{obs}) - \text{distance}(\text{traj} \leftrightarrow \text{traj})$$

$$ES(P, \mathbf{y}) = \underbrace{\frac{1}{N} \sum_{i=1}^N \sqrt{\sum_{m=1}^M \left(x_m^{(i)} - y_m \right)^2}}_{\text{distance from surveillance data}} - \underbrace{\frac{1}{2N^2} \sum_{i=1}^N \sum_{j=1}^N \sqrt{\sum_{m=1}^M \left(x_m^{(i)} - x_m^{(j)} \right)^2}}_{\text{distance between trajectories}},$$

Application: Flu Scenario Modeling Hub

- Strictly proper $\rightarrow$ better discrimination.
- Handles multi-peak dynamics.
- Integrates multiple targets in one metric.



A strictly proper scoring rule for evaluating epidemic model trajectories, capturing full temporal structure and enabling comprehensive model comparison.

Declarations of interest: none